Chapter 10

The Triumph of Utilitarianism: The Economics of Jevons, Menger, and Walras

The period from the mid-1840s to 1873 (the year that marked the beginning of the Long Depression in Europe) was one of rapid economic expansion throughout most of Europe. Industrialization was taking place in continental Europe as well as in the United States. England experienced a surge of industrial growth, with heavy industry, and particularly the capital goods industry, leading the way.

In the entire North Atlantic capitalist sphere, this rapid industrial growth was accompanied by an ever increasing degree of concentration of capital, industrial power, and wealth. In some cases, this growing concentration was the result of aggressive and destructive competition that eliminated small or weak competitors. In other cases, large, powerful competitors, seeing the potentially mutual destructiveness of such ruthless rivalry, combined in cartels, trusts, and mergers in order to ensure their survival.

During this period, there were also revolutionary changes in transportation and communication. These changes also furthered industrial concentration, because they made it possible for ever-widening markets to be supplied efficiently by a small number of giant companies or corporations. The joint-stock company, or corporation, became an effective means by which a single business organization could gain control over vast amounts of capital. A large, well-organized money market evolved in Europe and North America that successfully channeled the smaller capital holdings of thousands of individuals and small businesses into the hands of large corporations.

Thus, by the early 1870s, capitalism was beginning to take on a modified form—an economic system dominated by anywhere from several hundred to a thousand or so colossal corporations in the important spheres of industry, finance, transportation, and marketing. Although this concentration was to become significantly more severe by the early twentieth century, the new form of capitalism was emerging quite clearly by the 1870s.

Social relations among people in this new form of capitalism began to assume two distinct and drastically different forms. Within the giant corporation, social relations took on a hierarchical, bureaucratic form. Corporations were pyramidal social organizations in which each stratum was rigidly controlled and coordinated by the stratum above it. The entire pyramid was controlled from the top by a small group of owners or managers. All individual actions and economic or production processes within the corporation were integrated and coordinated in a rational, calculated manner. Elaborate systems of cost accounting, quality control, and scientific management were among the methods by which this rational control was exercised. The objective of the capitalists, of course, remained unchanged: to maximize profit and to accumulate more capital.

In its relationship to the remainder of the capitalist economy, however, the corporation was in no different a position than had been the smaller, individual capitalist of earlier decades. Among the giant corporations, the thousands of much smaller and less important business enterprises, and the tens of millions of workers, the older social relations of commodity production still prevailed. A massive, intricate system of mutual interdependence among all of these institutions and individuals continued to be mediated only by the blind, impersonal forces of the market.

Within this historical context, it might seem that economists would abandon Adam Smith's "invisible hand" conception of a capitalist economy, which had been based on the analysis of an economy composed of many small enterprises. In such an economy, no individual enterprise could exercise a significant influence on the overall market. The actions of any firm were dictated by consumer tastes, as registered in the marketplace, and by the competition of innumerable other small firms, each vying for consumers' dollars.

We have seen that this view of a capitalist economy combined quite naturally with the individualism and moral hedonism of utilitarianism to create the conclusion that capitalism was an economic system in which social harmony naturally prevailed. While the writings of Smith and Ricardo combined this view with a labor theory of value perspective that led to quite different conclusions, Bentham, Say, Senior, and Bastiat had gone far in freeing the individualistic, utilitarian perspective from the "dangerous," class conflict conclusions of the labor theory perspective.

If one believed that the history of economic theories should reveal nothing but a continuous and progressively closer approximation of economic theories to the existing reality of the evolving capitalist system, then it would appear that one would expect the individualistic, utilitarian perspective of Say, Senior, and Bastiat to be slowly replaced by one that reflected the changing economic and social forms of capitalism.

Such was not the case. During the early 1870s, at precisely the time when the drive toward the economic concentration of corporate capitalism was gaining momentum, three very famous economics texts were published. Wil-

liam Stanley Jevons's *The Theory of Political Economy*[1] and Carl Menger's *Grundsatze der Volkswirtschaftslebre* (*Principles of Economics*)[2] both appeared in 1871, and three years later Leon Walras's *Elements d'economie politique pure* (*Elements of Pure Economics*) was published.[3] Although there were many differences in the analyses of these men, the similarities in both the approaches and contents of these books were striking.

Each represented a continuation of the individualistic, utilitarian perspective of Say, Senior, and Bastiat. Each independently worked out a logically consistent solution to the water–diamond paradox that had led Smith to the conclusion that no direct relation between utility and exchange value could be found. This paradox had never been adequately resolved by Say, Senior, or Bastiat, even though these three thinkers had insisted that utility was the determinant of exchange value.

Jevons, Menger, and Walras formulated the version of the utility theory of value that remains at the heart of neoclassical orthodoxy to this day. They also added certain refinements and extensions to the ideas of Say, Senior, and Bastiat—the most important extension being Walras's conception of general economic equilibrium, which constituted one of the most important conceptual or analytical developments in the history of economic thought, and of which more will be said below. It is unquestionably because they were the first thinkers to provide a consistent theory of value within the general utilitarian philosophical perspective that later conservative economists have referred to their theories as a "revolution" in economic thinking, and have referred to the 1870s as a watershed separating old-fashioned classical economics from modern, scientific neoclassical economics.

The significance of the "marginalism" that was introduced into economic theory by these three thinkers would seem to depend on the historian's view of contemporary economic theory. A segment of the academic economics profession has evolved that appears to put great emphasis on the logical and mathematical rigor of a theory, quite independently of its content or the practical importance of its conclusions. Among economists of this bent, a theorist is admired to the extent that he or she can cast his or her theory in the most esoteric, complex, and rigorously mathematical form.

The notion of declining marginal utility (developed independently by each of the thinkers discussed in this chapter) permitted Jevons, Menger, and Walras and their successors to show concretely and explicitly how utility determined values (something that Bentham, Say, Senior, and Bastiat believed but could not demonstrate). Although this was definitely a major improvement over the doctrines of the earlier utility theorists, the major significance of Jevons, Menger, and Walras lay in how they changed the form of utilitarian economics and did *not* make any major change in its content. Marginalism permitted the utilitarian vision of human nature, which was considered to consist exclusively of the rational, calculating maximization of utility, to be formulated in terms of differential calculus. This then constituted the real beginning of the trend

toward esoteric mathematical formulation of economic theories (although Menger himself disliked the use of mathematics to express economic theory). It is probably for this reason that economists who value mathematical rigor for its own sake see Jevons and Walras as being the most important forebears of modern economic theory.

While these three thinkers' development of the principle of marginal utility unquestionably furnished an important link in completing the chain of the arguments of Say, Senior, and Bastiat, it hardly seems significant enough to view their ideas either as revolutionary (as some refer to this time as the beginning of the "marginalist revolution" in economics) or as a watershed dividing the history of economic theories. Their corrections to the formulation of the already widely held utilitarian notion that utility determines prices seem comparable in importance to those of later theorists who successfully worked out Marx's "transformation problem." Beyond this refinement of utilitarian price theory, only Walras—in his theory of general economic equilibrium—appears to have formulated a truly significant addition to the utilitarian tradition in economics (although the Physiocrats and Say had developed simpler versions of a general economic equilibrium, Walras went far beyond their sketchy, suggestive treatments).

Jevons's Theory of Marginal Utility and Exchange

William Stanley Jevons (1835–1882) wrote on a wide variety of topics, ranging from meteorology to logic to economic theory. His *Theory of Political Economy* was his most important work in the last field. In 1860, in a letter to his brother, he wrote: "in the last few months I have fortunately struck out what I have no doubt is the *true Theory of Economy,* so thorough-going and consistent that I cannot read other books on the subject without indignation."[4] This "true theory" was finally published in 1871.

In the preface to the *Theory,* Jevons stated that "Bentham's ideas . . . are . . . the starting point of the theory given in this work."[5] He had no doubt that utilitarianism was the only possible foundation for scientific economic theory: "In this work I have attempted to treat economy as a calculus of pleasure and pain, and have sketched out . . . the form which the science . . . must ultimately take."[6] The ultimate truth, on the basis of which he had felt such indignation toward other theories, was "*that value depends entirely upon utility.*"[7]

When Jevons used the term *value,* he always meant simply exchange value, or price. Whereas labor theorists, such as Marx, had defined value as the labor embodied in a commodity, Jevons contemptuously rejected such a definition:

> A student of economics has no hope of ever being clear and correct in his ideas of the science if he thinks of value as at all a *thing* or an *object,* or even as anything which lies in a thing or object. . . . The word value, so far as it can be correctly used, merely expresses *the circumstance of its exchanging in a certain ratio for some other substance.*[8]

Thus, Jevons was interested only in prices. He avowedly and proudly restricted his economic analysis to the sphere of circulation, the market. As Marx had pointed out some years before, within the sphere of the market all people are essentially identical. When Jevons wrote of people, he assiduously avoided any real discussion of superordinate or subordinate social relations. People, in Jevons's view, had only two characteristics that defined them as economic agents; moreover, every person had both of these characteristics. Hence, there was an abstract, implicit equality among all persons. The first characteristic was that they derived utility from consuming commodities: "Anything which an individual is found to desire . . . must be assumed to possess for him utility. In the science of economics we treat men not as they ought to be, but as they are."[9] The second characteristic was that every person was a rational, calculating maximizer. And rational, calculating, maximizing behavior was the only element of human action to be studied by economics: "To satisfy our wants to the utmost with the least effort—to procure the greatest amount of what is desirable at the expense of the least that is undesirable—in other words, to *maximize pleasure,* is the problem of economics."[10]

People received utility from consuming commodities. The error of previous economists, Jevons believed, lay in their failure to distinguish between the *total utility* a person derived from the consumption of a given quantity of some commodity and the "final degree of utility" (or, what in later neoclassical terminology came to be called the "marginal utility") that the person derived from consuming the *last small increment* of that commodity. Although it was often true that total utility might continue to become greater as one consumed an increased quantity of a commodity, the final "degree of utility . . . ultimately decreases as that quantity increases."[11] It was this "final degree of utility," or marginal utility, that concerned Jevons. This principle of diminishing marginal utility was to become the cornerstone of the neoclassical restatement of utilitarianism.

By introducing the notion of marginalism into utilitarian economics, Jevons had found a way in which the utilitarian view of human beings as rational, calculating maximizers could be put into mathematical terms. If the total utility one received from consuming a commodity depended on the quantity consumed, then this could be written as a mathematical function, $TU = f(Q)$, which simply says total utility (TU) has some concrete mathematical relationship to the quantity (Q) consumed. In calculus, the first derivative of a function tells one how much the dependent variable (in this case, total utility) changes as a consequence of an infinitesimally small change in the independent variable (in this case, the quantity consumed). The first derivative of the total utility function gives one the *marginal utility* at any particular quantity consumed. The logic of maximization could be easily formulated in terms of calculus. The total utility function was maximized when the quantity was increased to that point at which marginal utility was equal to zero. This was not very profound. It simply meant that to maximize one's utility from consuming a

particular commodity, one should consume the commodity (if there were no costs of consumption) until one was satiated, that is, until one could derive no more utility from another small increment of the commodity.

When consumption involved costs, those costs could be stated in mathematical form. For example, if one possessed a commodity y and could only get another commodity x by giving up some y in exchange, then one could compare the ratios of one's marginal utilities for the two commodities, MU_x / MU_y and the prices of the two commodities, P_x / P_y. If MU_x / MU_y was higher than P_x / P_y, then our individual could gain utility by trading some of his y for some x. If the process continued until the individual had exhausted the gains from exchange, he or she would have traded to the point at which $MU_x / MU_y = P_x / P_y$. To put the same thing differently, the ratio MU_x / P_x would tell one how much additional utility one would get (or give up) if one purchased (or sold) an additional dollar's worth of commodity x. The two individuals would, Jevons asserted, buy and sell the commodities until the marginal utilities of each commodity had changed to the point where $MU_x / P_x = MU_y / P_y$. At that point, the last dollar's worth of either x or y yielded the same increment to the individual's total utility. If for an individual the ratio MU_x / P_x was greater than MU_y / P_y, then that individual would sell y and buy x, thereby losing less utility for giving up a dollar's worth of y than he or she gained from the additional dollar's worth of x. But as he or she gave up y and gained x, the principle of diminishing marginal utility meant that MU_y would increase and MU_x would decrease until $MU_x / P_x = MU_y / P_y$. At that point, no further gains from exchange could be realized. The reverse but identical process would have occurred if MU_y / P_y had been greater than MU_x / P_x.

All prior utilitarian theorists had realized that in voluntary exchange, an individual bought or sold as long as what he or she purchased gave more utility than the utility lost in what he or she sold. This had always been the basis for advocating free exchange and for the belief that exchange harmonized everyone's interests. Jevons's only addition to the theory had been to give this principle a mathematical formulation and to make explicit the distinction between total utility and marginal utility.[12] Thus Jevons's major addition to the ideas of the previous utilitarian economists can be summed up in his own words: "The nature of wealth and value is explained by the consideration of indefinitely small amounts of pleasure and pain."[13] "I contend that all economic writers must be mathematical so far as they are scientific at all."[14]

Jevons attempted to show how marginal utility determined prices. But in doing so he tried to show how two "trading bodies" could arrive at equilibrium prices for two commodities. But the theoretical problem as he set it up did not yield any determinate solution, and it was left to other neoclassical economists to demonstrate how the marginal-utility theory could become a theory of prices. Jevons simply demonstrated how consumers arranged their exchanges, once prices were known, in order to maximize their individual utilities.

Jevons did not, however, derive from the Benthamite notion of diminish-

ing marginal utility the egalitarian conclusions of Bentham. In our critique of Thompson's utilitarianism, we argued that interpersonal comparisons of the relative intensities of the utility of wealth or income were impossible because pleasure was a purely subjective, personal experience. From this we concluded that utilitarianism could ethically compare two social situations only when unanimity prevailed among all participants. This, we argued, gave utilitarianism its highly conservative bent. The situations before and after an exchange are among the few situations in any social setting where such unanimity prevailed. If both parties participated in the exchange voluntarily, then it could be assumed that both benefited. This seemingly trivial conclusion has always been the basis of the utilitarian belief in the natural harmony of market capitalism.

Unlike Bentham, Thompson, and Mill, Jevons clearly recognized (and appreciated) this limitation of utilitarianism:

> The reader will find . . . that there is never, in any single instance, an attempt made to compare the amount of feeling in one mind with that in another. I see no means by which such comparison can be accomplished. The susceptibility of one mind may, for what we know, be a thousand times greater than that of another. . . . Every mind is thus inscrutable to every other mind, and no common denominator of feeling seems to be possible.[15]

Not surprisingly, Jevons felt certain that social harmony and not class conflict was the natural state of market capitalism. He asserted that "the supposed conflict of labour with capital is a delusion."[16] Appealing to universal brotherhood, he added: "We ought not look at such subjects from a class point of view," because "in economics at any rate [we] *should regard all men as brothers.*"[17] This "brotherhood" of social harmony arose, of course, because all people appear essentially equal and in the same light when seen exclusively as exchangers:

> Each labourer must be regarded, like each landowner and each capitalist, as bringing into the common stock one part of the component elements, bargaining for the best share of the produce which the conditions of the market allow him to claim successfully.[18]

> He who pays a high price must either have a very great need of that which he buys, or very little need of that which he pays for it; on either supposition there is gain by exchange. In questions of this sort there is but one rule which can be safely laid down, namely that no one will buy a thing unless he expects advantage from the purchase; *and perfect freedom of exchange, therefore, tends to the maximizing of utility.*[19]

Again the utility perspective had arrived at a new appreciation of the "invisible hand" that now, with Jevons's new "scientific" and "mathematical" formulation, could be shown to maximize everyone's utility in a world of brotherhood and harmony.

Jevons also developed a theory of capital that, like those of Ricardo and

Marx, stressed the temporal dimension of production. A highly similar theory of capital was to become central to the later Austrian and Chicago schools of neoclassical economics (both of which grew out of the influence of Menger and reflected the spirit of Bastiat). This capital theory was developed only a few years later by a disciple of Menger—Eugen von Böhm-Bawerk. Because Böhm-Bawerk's version was superior to that of Jevons, and because the theory is usually associated with Böhm-Bawerk, we will postpone a discussion until we consider his ideas in the next chapter.

The principal objective of Jevons's capital theory was to refute Ricardo's conclusion that the rate of profit varied inversely to the rate of wages. Ricardo's conclusion obviously demonstrated the fundamental antagonism between capital and labor, and Jevons did not like it. Discussing Ricardo's theory, Jevons wrote:

> We thus arrive at the simple equation—
>
> produce = profit + wages
>
> A plain result also is drawn from the formula; for we are told that if wages rise profits must fall, and vice versa. But such a doctrine is radically fallacious. . . . The wages of a working man are ultimately coincident with what he produces, after the deduction of rent, taxes and the interest of capital.[20]
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> It is the proper function of capitalists to sustain labour before the result is accomplished, and as many branches of industry require a large outlay long previous to any definite result being arrived at, it follows that capitalists must undertake the risk of any branch of industry where the ultimate profits are not accurately known. . . . The amount of capital will depend upon the amount of anticipated profits, and the competition to obtain the proper workmen will strongly tend to secure to the latter all their legitimate share in the ultimate produce.[21]

Since competition would secure to the worker his "legitimate share," Jevons hoped the trade unionist, who saw the capitalist as a class enemy, would "cease his exclusive strife against his true ally, his wealthy employer."[22] Because the accumulation of capital benefited all workers, Jevons believed that the laborer should view the capitalists as "the trustee who holds his capital rather for the good of others than himself."[23] The common interest of both the wealthy "trustee" and the laborer who "benefited" from the trustee's wealth were both promoted by free exchange. Bastiat had asserted that economics was exchange—pure and simple. Jevons wrote:

> Exchange is so important a process in the maximizing of utility . . . that some economists have regarded their science as treating of this operation alone. . . . I am perfectly willing to agree with the high importance attributed to exchange. It is impossible to have a correct idea of the science of economics without a perfect comprehension of the theory of exchange.[24]

Jevons, however, did not want economics to be seen as explaining only exchange. Both Thompson and Hodgskin had argued, on utilitarian grounds, that exchange would be even more beneficial in an economy where workers owned

their own means of production. Jevons did not want anyone to forget that capitalists' ownership of capital was sacrosanct and that "it is the proper function of capitalists to sustain labour."[25] Therefore, he extended Bastiat's definition of economics (in a way with which Bastiat surely would have agreed): "Economics, then, is not solely the science of exchange or value: it is also the science of capitalization."[26]

It is not surprising that Jevons's indignation toward prior economists, which he had expressed to his brother in the letter written in 1860, did not extend to all prior economists, but was directed primarily toward Ricardo and Mill:

> When at length a true system of economics comes to be established, it will be seen that that able but wrong-headed man, David Ricardo, shunted the car of economic science on to a wrong line—a line, however, on which it was further urged toward confusion by his equally able and wrong-headed admirer, John Stuart Mill. There were economists, such as Malthus and Senior, who had a far better comprehension of the true doctrines.[27]

Jevons's *The Theory of Political Economy* is replete with condemnations of Ricardo and Mill and laudatory statements describing the doctrines of Malthus, Say, Senior, and Bastiat. "J.B. Say has correctly . . . defined utility,"[28] he wrote, and the doctrine was correctly developed by Senior "in his admirable treatise" and by "Bastiat, for instance, in his *Harmonies of Political Economy.*"[29] In another essay, while discussing Malthus's theory of population, Jevons referred to Malthus as "one of the most humane and excellent of men."[30]

One of John Maynard Keynes's favorite quotations from Malthus's writings shows quite clearly what type of man Jevons considered to be "most humane and excellent." The quotation is from Malthus's *Essay on Population,* and it was in the context of discussing Malthus's population theory that Jevons expressed his admiration of Malthus.

> A man who is born into a world already possessed, if he cannot get subsistence from his parents on whom he has a just demand, and if the society does not want his labour, has no claim of *right* to the smallest portion of food, and, in fact, has no business to be where he is. At nature's mighty feast there is no vacant cover for him. She tells him to be gone, and will quickly execute her own orders, if he does not work upon the compassion of some of her guests. If these guests get up and make room for him, other intruders immediately appear demanding the same favour. The report of a provision for all that come, fills the hall with numerous claimants. The order and harmony of the feast is disturbed, the plenty that before reigned is changed into scarcity; and the happiness of the guests is destroyed by the spectacle of misery and dependence in every part of the hall, and by the clamorous importunity of those, who are justly enraged at not finding the provision which they had been taught to expect. The guests learn too late their error, in counteracting those strict orders to all intruders, issued by the great mistress of the feast, who, wishing that all her guests should have plenty, and knowing that she could not provide for unlimited numbers, humanely refused to admit fresh comers when her table already was full.[31]

One wonders what a person would have to have done or said for Jevons to have considered that person inhumane.

Rereading the doctrines of the four economists whom Jevons most admired, it strikes one as rather strange that Jevons's ideas are frequently referred to as constituting a revolution in economic theory that marks a watershed dividing older from more modern views. The fundamental differences between the utility-theory perspective and the labor-theory perspective were clear before Jevons wrote anything, and his contribution was primarily to show that marginalism permitted the doctrines of Malthus, Say, Senior, and Bastiat to be stated with mathematical elegance and greater logical consistency. But the theoretical and ideological essence of the utility perspective remained unchanged.

Menger's Theory of Marginal Utility, Prices, and Income Distribution

Carl Menger (1840–1921) achieved eminence through his writings both in economic theory (the *Principles of Economics* being his principal work) and in methodology (some of his writings on methodology exist in English translation under the title *Problems of Economics and Sociology*).[32] In his economic theory, he rejected the use of mathematical equations and expressed his theories verbally with the aid of numerical examples.

His description of total utility and marginal utility was similar to that of Jevons. He illustrated the principle with a table of numbers. Table 10.1 reproduces the numbers from Menger's table with some labels added for clarification.[33] In the table, to find the marginal utility of some commodity (say type II) one goes to the second column and then reads down to the number of units consumed. If six units of commodity II were consumed, for example, the marginal utility of the sixth unit would be 4. The total utility (not illustrated in the table) can be calculated simply by summing the marginal utilities down to the number of units consumed. For example, six units of commodity II yield a total utility of 39.

The relationship between total utility and marginal utility can be graphed (assuming that the units of commodities can be subdivided and therefore that the lines relating utility to quantities consumed are smooth and continuous). Figure 10.1 illustrates the relationship between the marginal utility and the total utility of commodity II in Menger's table. This general relationship between total quantities and marginal quantities recurs in neoclassical economics, and several types of maximizing problems can be analyzed in a similar manner.

Menger used his table to illustrate how a consumer maximized his or her utility. We quote Menger's own explanation of the table:

> Suppose that the scale in column I expresses the importance to some one individual of satisfaction of his need for food, this importance diminishing according to the degree of satisfaction already attained, and that the scale in column V expresses similarly the importance of his need for tobacco. It is evident that satisfaction of his need for food, up to a certain degree of completeness, has a decidedly higher importance to this individual than satisfaction of his need for tobacco. But if his need for food is already satisfied up to a certain degree of completeness (e.g., if a further satisfaction of his need for food has only the importance to him that we

Table 10.1

An Illustration of Declining Marginal Utility

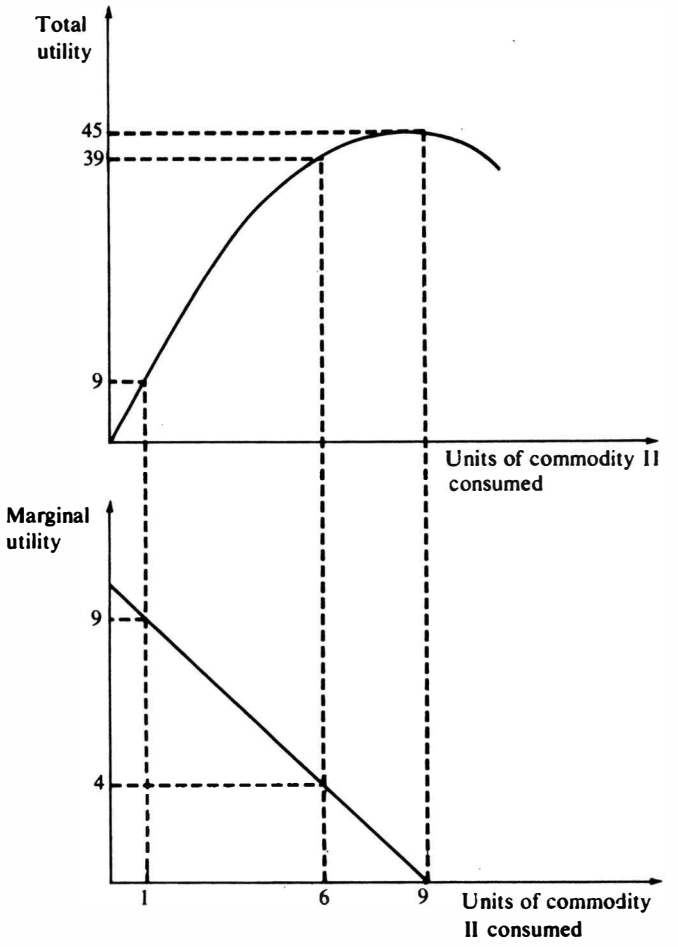
Number of units consumed	Type of commodity									
	I	II	III	IV	V	VI	VII	VIII	IX	X
1	10	9	8	7	6	5	4	3	2	1
2	9	8	7	6	5	4	3	2	1	0
3	8	7	6	5	4	3	2	1	0	
4	7	6	5	4	3	2	1	0		
5	6	5	4	3	2	1	0			
6	5	4	3	2	1	0				
7	4	3	2	1	0					
8	3	2	1	0						
9	2	1	0							
10	1	0								
11	0									

> designated numerically by the figure 6), consumption of tobacco begins to have the same importance to him as further satisfaction of his need for food. The individual will therefore endeavor, from this point on, to bring the satisfaction of his need for tobacco into equilibrium with the satisfaction of his need for food.[34]

Menger believed that the equilibrium at which the individual maximized his or her utility was achieved when the individual equated the marginal utility derived from any one commodity to the marginal utility derived from each of the other commodities he or she consumed. This statement of the maximizing condition was inferior to that of Jevons because Menger neglected prices. Menger's maximizing solution would hold only if the price per unit of each of his types of commodities were equal to the price per unit of every other type. In that case Jevons's equation $MU_x / P_x = MU_y / P_y$ would have the same number in the denominator of each ratio, and the equality of the ratios would require that the numerators (or the marginal utilities) be equal. Jevons's formulation was the correct one, and Menger's formulation was, although he did not realize it, a special, highly unlikely case.

Menger's discussion of price determination, however, was superior to Jevons's. Menger explained prices on the basis of supply and demand. All of the classical economists had explained short-run market prices on the basis of supply and demand. So, in this respect, the classical, Marxist, and neoclassical economists have never differed. The differences have been in their explanations of what underlies supply and demand. Smith, Ricardo, and Marx sought to find explanations of rent, wages, and profit that were outside the realm of prices.

Figure 10.1 **Relationship of Total Utility and Marginal Utility**



Rent, wages, and profits were both the components in the class distribution of income for the total society and the components of cost for an individual firm's output. As cost components, their summation was Smith's "natural price" or Marx's "price of production" of a commodity. When an industry was in equilibrium, the market price, as determined by supply and demand, would equal the natural price, or price of production. This was the nature of the pricing process when viewed from the labor-theory perspective—the distribution of income was independent of prices, and, in fact, determined them.

The utility perspective viewed the pricing process in an entirely different manner: supply and demand determined prices, and supply and demand were, in turn, explained by utility. Therefore, utility was the ultimate determinant of the prices of consumer goods. The prices of the "factors of production"—land, labor, and capital—were also determined by their supply and demand. Their supply was determined by the calculations of utility on the part of their owners,

and their demand by *their productivity* in producing consumer goods as well as by the utility consumers derive from consuming these goods. Thus, in the utility perspective, wages, rent, and profit were at least partially determined by the prices of consumer goods. The difference in the two views of the pricing process was indeed distinct. Menger's superiority over Jevons lay in the way in which he described this pricing process from his utility-theory perspective.

Menger explained demand for consumer goods by showing that when a particular good had a high price in relation to the marginal utility that most consumers could get from that commodity, most consumers would get more utility by keeping their money than by spending it on that good. But as the price of the good dropped, more consumers would find that the marginal utility derived from consuming that commodity exceeded the utility that they lost by giving up the smaller amount of money. Moreover, as the price dropped, consumers who were already consuming the good would find that utility maximization required them to buy more of the good. Thus, from Menger's principle of diminishing marginal utility, he derived the law of demand: The quantity of a commodity that people were willing to purchase depended on the price of the commodity, and the quantity demanded and the price were inversely related to each other.[35]

Menger's discussion of supply was less adequate. He usually treated supply as a preexisting quantity already in the hands of the seller. The seller, using utility maximization as a guide, decided what quantities he or she wished to sell at any given price. The combination of the desires to buy and sell (all determined by utility considerations) determined prices. Menger went on to show that a monopolist selling a commodity would charge higher prices and sell smaller quantities than would have obtained if the commodity had been sold by many competitive sellers.[36] Menger, therefore, extolled the benefits of free competition.

Menger called commodities produced for consumption "goods of first order," and the factors of production he called "goods of higher order."[37] He wished to demonstrate two relationships between these two kinds of goods. First, a variety of goods of higher order was necessary to produce "goods of lower order," and, therefore, capital and land were as necessary to production as was labor.[38] In this assertion, as we have seen, he did not differ from the labor theory of value. Second, he wanted to show, in direct contradiction to the labor theory, that "the value of goods of higher order is always and without exception determined by the prospective value of the goods of lower order in whose production they serve."[39] In this regard, he considered the labor theory of value to be "among the most egregious of the fundamental errors that have had the most far-reaching consequences in the previous development of our science."[40]

Menger included labor, raw materials, and tools as goods of higher order. Like Jevons, his conception of capital involved more than simply the tools and materials of production; it focused primarily on the time dimension of

production. He argued that the "satisfaction of our needs is . . . dependent on the command of quantities of economic goods for certain periods of time (on capital services). . . . For this reason, capital services are objects to which we attribute value."[41] Because Menger believed that no good of higher order could produce a consumption good by itself—that is, that goods of higher order were complementary and had to be used together—he had a difficult time demonstrating how one could measure the productivity of a single good of higher order. Like nearly every thinker in the utility tradition, he believed that the price paid to each individual productive input reflected the size of each input's productive contribution. Measuring that productive contribution was, therefore, very important.

Menger's solution to this problem of measurement was summarized in this way:

> Assuming in each instance that all available goods of a higher order are employed in the most economic fashion, the value of a concrete quantity of a good of higher order is equal to the difference in importance between the satisfactions that can be attained when we have command of the given quantity of the good of higher order whose value we wish to determine and the satisfactions that would be attained if we did not have this quantity at our command.[42]

Menger's analysis was suggestive but inadequate. If, for example, the firm being considered was a farm, and the good of higher order being considered was seed grain, then without the seed grain, production would be impossible. Using his method, Menger would have to conclude that the seed grain produced all of the value of the output, despite the use of land, tractors, fertilizers, and so on. Needless to say, this was not Menger's intention. A consistent theory of input pricing within the utility perspective required that inputs be at least partially substitutable and that one examine the effect on output of a small marginal increment of one input. This, in fact, was the method adopted in the marginal-productivity theory, which we will discuss in the next chapter.

Having established that inputs were paid according to their productivity, Menger made an assertion that was to characterize most versions of marginal utility economics: When each input was paid the value of its productive contribution, the value of the total output that had been produced would be exactly exhausted. Therefore, *there was no surplus for any person or class to expropriate.* The economic, social, political, and ideological implications of this assertion—antithetical to the doctrines of Smith, Ricardo, Thompson, Hodgskin, and Marx—are obvious. In Menger's words:

> The aggregate present value of all the complementary quantities of goods of a higher order (that is, all the raw materials, labor services, services of land, machines, tools, etc.) necessary for the production of a good of . . . first order is equal to the prospective value of the product. But it is necessary to include in the sum not only the goods of higher order technically required for its production but also the services of capital and the activity of the entrepreneur. For these are as unavoidably necessary in every economic production of goods as the technical requisites already mentioned.[43]

Menger felt no need to justify profits by claiming that capitalists engaged in painful abstemious behavior. He simply stated that "the harmony of the needs that the individual households attempt to satisfy is reflected in their property."[44] Then he reassuringly affirmed that "the entire sum of goods at an economizing individual's command for the satisfaction of his needs, we call his *property.* His property is not, however, an arbitrarily combined quantity of goods, but a direct reflection of his needs."[45] He continued:

> Human economy and property have a joint economic origin since both have, as the ultimate reason for their existence, the fact that goods exist whose available quantities are smaller than the requirements of men. Property, therefore, like human economy, is not an arbitrary invention but rather the only practically possible solution of the problem that is, in the nature of things, imposed upon us by the disparity between requirements for, and available quantities of, all economic goods.[46]

We will see in the next section that Menger believed society was an organic whole that had evolved with an inherent, deterministic necessity to its existing state. Social institutions and laws, he believed, could not and should not be tampered with. Thus, the necessity of property was, for him, a necessity of the particular property relations of existing capitalist society.

After developing this theory of income distribution (or input pricing) and making these assertions about private property, Menger felt no need to justify rent and profit (or interest, which he conceived to be the total return to the ownership of capital). Such income was *absolutely necessary* and *absolutely inevitable,* he believed. Why condemn or justify the necessary and the inevitable? Just accept its inevitability and be happy about the harmony existing in the situation:

> One of the strangest questions ever made the subject of scientific debate is whether rent and interest are justified from an ethical point of view or whether they are "immoral." . . . Wherever the services of land and of capital bear a price, it is always as a consequence of their value, and their value to men is not the result of arbitrary judgments, but a necessary consequence of their economic character. The prices of these goods (the services of land and of capital) are therefore the necessary products of the economic situation under which they arise, and will be more certainly obtained the more developed the legal system of a people and the more upright its public morals.[47]

Menger, of course, was doing everything in his power to give a scientific foundation to the efforts to create and maintain "upright . . . public morals."

Menger's Arguments on Methodology

Menger was one of the most important participants in an extensive debate over the proper methodology for the social sciences. The debate occupied the attention of many of the leading German-speaking intellectuals in the late nineteenth century and has come to be known as the *Methodenstreit.* We will not attempt

to summarize the issues involved in the debate; instead, we will merely discuss two of the central assertions that Menger attempted to defend.

His first assertion was that "pure science" was always value free. Normative, moral, or ethical values were, he believed, completely foreign to science. In their values, individuals were influenced by their personal circumstances, class position, and emotions, and hence widespread agreement was never very likely on ethical questions. Science, however, was the description and understanding of actual reality—not reality as one wished it to be. Therefore, to the degree that scientists purged their theories of values, they could eventually come to complete and universal agreement.

Economics, then, to the extent that it was a science, was value free. Menger believed that many of the confusions in the writings of earlier economists had been the result of their failure to understand that ethical judgments had no place in pure economic theory. "The so-called 'ethical orientation' of political economy," he wrote, "is thus a vague postulate devoid of any deeper meaning both in respect to the theoretical and the practical problems of the latter, a confusion in thought."[48]

This, of course, was a repetition of Nassau Senior's methodological assertion. In our discussion of Senior, we argued that all theorizing—especially in the social sciences—is based on value judgments. Senior honored his own methodological argument only in the breach; all of his writings were intended to persuade the reader of his correctness on important ethical issues. While Menger and the many modern neoclassical economists who have expressed reverence for Senior's methodological argument have been considerably more subtle in their pervasive and persistent violations of the principle, these violations are nevertheless relatively easy to uncover in their writings. Menger's attempt to put the laws of private property and the distribution of income above either theoretical or moral dispute constitutes a most important violation of the principle of value-free social science.

Menger's second methodological principle was that economists could scientifically understand only individual households or business firms (Menger called these "individual economies"). They could never develop a similar scientific understanding of social aggregates such as classes or nations. Menger disliked the term "national interest," which was so frequently used to justify proposals for national reform, because he believed that its use was based on the false belief that a national economy with its own interests, separate and distinct from "individual economies" and their individual interests, could be identified and studied. The error of these reformers, he insisted, was that they saw the nation "itself as a large individual economy in which the 'nation' is to represent the needing, economic and consuming subject."[49] The "scientific" doctrine was then juxtaposed to this erroneous view:

> That phenomenon, which is commonly designated by "national economy," always presents itself to us, rather, merely as an organized complex of individual economies, as a multiplicity of economies joined together into a higher unity which is nevertheless not an economy itself in the strict meaning of the word.[50]

In our discussions of Bentham, Thompson, and Mill, we argued that utilitarianism has an extreme individualist orientation. As a result, moving from the notion that individual desires are the only criteria of good and bad to the notion that the social welfare is promoted by policies that maximize the total of all pleasures, entails innumerable logical and practical difficulties. Menger correctly perceived that reform based on utilitarian principles was inconsistent with utilitarian individualism. He therefore rejected the notion that there could possibly be such a thing as total national pleasure that could be increased through reform.

But we have also seen that the social conservatism of the utility theory, or the market-exchange perspective in economic theory, has always been based on a belief or faith that the existing laws of property ownership are natural, eternal, or above question. Therefore, in order for Menger's individualism to culminate in the ethical conclusions that characterize all of his writings, he had to have some sort of moral defense of the existing laws of private property, which he did.

Although the national economy could not be scientifically understood in the same manner as individual economies, Menger insisted that we could have some form of understanding of it. The national economy, he asserted, was "an *organism of economies.*"[51] He praised the German "historical school of jurists" (as opposed to the German historical school of economists, whom he attacked), who had recognized that a nation and the laws that defined its particular institutional arrangements constituted

> an "organic" structure which cannot and must not be arbitrarily shaped by individuals or by single generations, that is a structure which, on the contrary, is opposed as something higher to the arbitrariness of the individual, of the entire age, of human wisdom. . . . [The] "subconscious wisdom" which is manifested in the political institutions that came about organically stands high above meddlesome human wisdom.[52]

Menger concluded that "law is *not a chance affair,* but, both in terms of its essential idea and its particular content, it is something *implicitly* given essentially by human nature and the particularity of conditions."[53]

Menger expressed regret that there had not evolved

> a historical school of economists comparable to the historical school of jurists, which would have defended existing economic schools and interests against the exaggerations of reform thought in the field of economy, but especially against socialism. [Such a school] would have fulfilled a certain mission in Germany and prevented many a later setback.[54]

Thus, we see that Menger's methodological individualism and his belief that his theories were value free led to the belief that existing institutions and laws were above reform; reform efforts were, in his opinion, unscientific and socially harmful. His individualism ended in an appreciation of the "benevo-

lence" of the "subconscious wisdom" of the "organic whole." His value-free "science" ended in an assertion of the moral importance of defending existing economic interests against the "exaggerations" of socialists.

It is not surprising, therefore, that in his *Principles of Economics,* after discussing what he believed to be the determinants of the distribution of income, Menger wrote:

> It may well appear deplorable to a lover of mankind that possession of capital or a piece of land often provides the owner a higher income for a given period of time than the income received by a laborer for the most strenuous activity during the same period. Yet the cause of this is not immoral.[55]

It is, of course, no less an ethical judgment to say that something is not immoral than to say that it is immoral.

The ethical orientation of all Menger's writings (as was the case for Senior and for most present-day neoclassical economists professing to have value-free theories) was revealed in his consistent defense of existing economic interests against reformers and socialists:

> The agitation of those who would like to see society allot a larger share of the available consumption goods to laborers than at present . . . [is based on] a view of providing them with a more comfortable standard of living and achieving a more equal distribution of consumption goods and of the burdens of life. A solution of the problem on this basis, however, would undoubtedly require a complete transformation of our social order.[56]

Given Menger's faith in the benevolence of the "subconscious wisdom" embodied in the existing social, economic, and legal institutions, such a social transformation was morally unthinkable.

Walras's Theory of General Economic Equilibrium

Walras's independent development of the theory of diminishing marginal utility (he used the term *rareté* for marginal utility) and his discussion of the equation that gave the maximum utility a consumer could achieve through exchange need not be summarized here, because we have examined similar contributions in the writings of Jevons and Menger. Walras's most important and lasting contribution to economic theory was his theory of general economic equilibrium.

Although most previous economic theorists had discussed interrelationships between different markets, no economist prior to Walras set out a general conceptual and theoretical structure within which to examine the multiplicity of relationships among different markets. He realized that the forces of supply and demand in any one particular market depended in varying degrees on the prices prevailing in innumerable other markets.

For example, the demand of a consumer for a particular good depended, as

Jevons and Menger had argued, on the marginal utilities the consumer would derive from consuming various quantities of that good and the price of that good. But the condition for utility maximization showed that the consumer would purchase that particular good up to the point at which the ratio of its marginal utility over its price was equal to the same ratio for all other goods available for purchase. Therefore, the consumer's demand for that good depended also on the prices of all other consumer goods.

Now, insofar as the price of any good was, for the utility theorists, determined solely by the supply and demand for that good, and insofar as the demand for any good was determined by *both* all consumers' utilities from consuming that good *and the prices* of all consumer goods, it followed that to determine the price of the good, the prices of all other goods would have to be known. But the demand for those other goods (and hence their price) also depended on the price of the good in question. Therefore, a general theory of price determination was required. In such a theory, all prices would have to be determined simultaneously by both the total of all consumer utilities and the interrelations that existed among all markets.

However, such interrelationships did not exist only in the demand for consumer goods. They also existed in the supply of consumer goods, as well as in the demands and supplies of the other types of commodities or assets that were exchanged. Walras attempted to formulate a general theoretical framework within which he could show how, through the interactions of all markets, all prices could be determined simultaneously.

The purely logical requirement for such a general equilibrium theory was (as in all theoretical systems where several unknown variables must be determined simultaneously) that the number of unknown variables must equal the number of independent equations designed to determine the variables. There are additional logical conditions that must be fulfilled to guarantee the existence of a solution to this system of simultaneous equations, to guarantee that this solution be meaningful—that is, that it not contain solutions such as negative prices, which contradict some of the premises on which the theory is constructed—and to guarantee that the solution is unique. Walras, however, was primarily concerned to show that a theoretical system could be constructed in which the number of independent equations was sufficient to solve for the number of unknowns.

But if this had been his only concern, he could simply have posited any number of equations (which are, after all, not difficult to concoct). But he wanted his equations to have economic meaning and to describe what he considered real market forces that were important in determining prices. Therefore, before going on to describe Walras's system of equations, we must discuss the notion of general-equilibrium analysis in economics.

Needless to say, no theory can explain everything simultaneously. This would require an omniscience by which the entire universe was completely understood. Obviously all "general" theories are partial theories in the sense

that there are innumerable phenomena that they do not purport to explain. The difference between general-equilibrium theory and partial-equilibrium theory in economics is that whereas the former usually attempts to explain all prices and quantities exchanged within an entire economy for a given period, the latter takes as *given* all prices and quantities exchanged except for one or two and attempts to explain those one or two markets within the context of these given prices and quantities. In the next chapter, we will discuss the partial-equilibrium theory of Alfred Marshall.

In Walras's general-equilibrium theory, all prices and quantities exchanged were to be explained. This meant, of course, that they had to be explained on the basis of their relationships to some other elements in the social and economic environment. There had to be, in his theory, a description of some features of the social and economic setting of the market situation that could be used to explain prices and quantities exchanged.

In his equations, therefore, prices and quantities exchanged would constitute the dependent variables, and those features of the socioeconomic setting would constitute the independent variables (which could then, of course, be seen as the determinants of the dependent variables). The precise form of the equations would reflect the real economic relationship that Walras believed existed between the given features of the socioeconomic setting (his independent variables) and the prices and quantities exchanged in the market (his dependent variables).

The institutional setting of Walras's theory was competitive capitalism in which "there are land-owners, workers, and capitalists."[57] These three classes functioned in two economically important ways: first, as owners of productive services (land, labor, and capital), they supplied these services in the market; second, as consumers, they demanded consumer goods in the market. Walras took the existing distribution of ownership of "productive services" (i.e., the existing class division of society) for granted—as have nearly all the economists in the utility tradition. Taking class distinctions—and hence, different people's various roles in the production process—for granted, the most important element within the socioeconomic context was, for Walras, people's *subjective desires,* or their schedules of marginal utilities.

There were three important institutional factors, therefore, in Walras's theory. The first was the acceptance of the existing laws of and distribution of property as morally right and just. Here he simply stated that "the *ownership* of property . . . is legalized appropriation, or appropriation in conformity with justice."[58] Although he sometimes added that he intended to show why existing property rights embodied justice,[59] he never did. Rather, he repeatedly asserted the justness of existing property relations. For example, eleven pages after the above quotation, he wrote: "Property consists in fair and rational appropriation, or rightful appropriation."[60] Second, he assumed that the economy was made up exclusively of small, relatively powerless business firms, and perfect competition prevailed. Although he realized that "the principle of free competition is

not generally applicable . . . [to] monopolies,"[61] he ignored monopolies entirely throughout his general equilibrium discussion. Only after this discussion, and in the next to last chapter of his book, did he devote a few pages to a discussion of monopolies. He never attempted to tie this latter discussion back into his discussion of general equilibrium. Third, people were simply assumed to have measurable marginal utility schedules. There was no discussion of how these utility schedules came into existence or how they changed over time. He simply assumed that during the time period under analysis "the utility . . . remains *fixed* for each party."[62] Thus, utility was a metaphysically given, ultimate datum, in terms of which everything was to be explained.

While Walras did admit that people's utility schedules would change over time, he simply assumed that when such changes occurred, one simply had the new but essentially identical problem of finding a general equilibrium solution for a new time period. This was much more than a mere analytical simplification. Walras's utilitarian ideology justifying market capitalism depended on his conclusion that equilibrium prices accurately reflected people's *needs* or utilities, and, thereby, human satisfaction was maximized.

The conservative ideology embodied in the utility perspective loses what force it has if one admits desires are either socially determined or in a state of constant flux. Either of these possibilities leads to the question of a higher standard by which to judge desires themselves—a question that utilitarianism never considers. It is therefore not surprising that Walras asserted that "any value in exchange, once established, partakes of the character of a natural phenomenon, natural in its origins, natural in its manifestations and natural in essence."[63] By contrast, the labor theory of value stresses the view that prices are *social.*

After assuming the existing distribution of ownership, a perfectly competitive market, and metaphysically given, fixed utility schedules, Walras developed his system of equations to show what quantities of the various commodities were exchanged and how their prices were determined. He posited n productive services, m consumer goods, n prices of productive services, and m–1 prices of consumer goods (one consumer good was taken as the *numeraire* so that its price was one, by definition). Therefore, there were $m + n$ quantities of productive services and consumer goods exchanged, and $m + n$–1 prices at which they were exchanged. The total number of dependent variables, then, was $2m + 2n$–1.

There were four sets of equations that Walras set up to solve for the magnitudes of his dependent variables.

In the first set, the supply of each of the n productive services depended on the prices of every productive service and every consumer good. Thus, n equations were derived, relating the quantity supplied of each productive service to all prices in the system. The particular mathematical form of each equation would depend on the marginal utility schedules of all owners of productive services. At any particular set of the $m + n$–1 prices, they could make all of the

appropriate utility calculations and decide exactly how much of their productive services to sell in order to buy consumer goods so as to maximize their utilities. Therefore, the owners of any given productive service would have a particular amount that they wished to supply at each and every possible set of prices. There would be a separate equation relating these quantities supplied of each of the n factors to all possible sets of prices.

In the second set, the demand for each of the m consumer goods depended on all of the $m + n-1$ prices. Thus, m equations were derived. The reasoning was identical to that in the first set of equations. Productive service owners sold their services in order to buy consumer goods, and utility maximization would dictate exactly how much of the services that they supplied and how much of each consumer good that they would buy for each and every possible set of all of the $m + n-1$ prices.

In the third set, in order for the economy to be in equilibrium, the demand for each productive service would have to equal the supply. Full employment of resources was assumed as a condition of equilibrium. This furnished one equation (equating supply and demand) for each of the n productive services.

In the fourth set, Walras believed that the assumption of perfect competition would assure that the price of each consumer good would have to equal its cost of production. The cost of production would depend on the "technical coefficients" of production (or technical recipes for production) and the prices of the productive services.

Therefore, Walras had $2m + 2n$ equations—m from each of the second and fourth sets of equations and n from each of the first and third sets of equations—to solve for $2m + 2n-1$ unknowns. In the next several paragraphs, we will discuss what has come to be known as "Walras's law," which proves that if all markets but one are in equilibrium, then the last market must also be in equilibrium. This means that one of Walras's equilibrium equations was not an independent equation, because if all of the other equations were simultaneously solved, it would automatically be solved also. Therefore, he ended up with $2m + 2n-1$ independent equations—the same number as the unknowns that had to be solved.

Walras's law is really a definitional identity. It states that with any given set of prices, the total demand for all things exchanged must equal the total supply of all things exchanged. It follows from the definition of supply and demand because, at some set of prices, the desire to exchange definitionally implies the desire to acquire something at those prices (demand) by giving up something (supply) of equal value. Therefore, every individual demand is simultaneously a supply of the same magnitude, and so if these individual demands and supplies are aggregated, the totals must be equal. Walras (using the word *offer* where we have used *supply*) stated the principle in this way:

> The effective demand for or offer of one commodity in exchange for another is equal respectively to the effective offer of or demand for the second commodity multiplied by its price in terms of the first. . . . Indeed, demand ought to be considered as the

> principal fact and offer as the accessory fact where two commodities are exchanged for each other in kind. No one ever makes an offer simply for the sake of offering. The only reason one offers anything is that one cannot demand anything without making an offer. Offer is only a consequence of demand.[64]

Walras's law is obviously true by definition. However, many defenders of Say's law have appeared to confuse it with Walras's law. Say's law implies that there will be a demand for all newly produced commodities. This does not follow from Walras's law. People might produce in order to exchange this output for some limited quantity of previously existing assets, such as money. There may not be as much money in existence as people desire to own at the existing set of prices. In such a case, there would be an excess supply of newly produced output, just matched by an excess demand for money. There would, therefore, be a general "glut" of commodities (to use Malthus's term), even though Walras's law still held. In fact, at any set of prices, Walras's law always holds even when *every single individual market is out of equilibrium.*

But Walras's law tells us something useful, even though it is merely definitional. It proves that in any disequilibrium, the total of excess demand (in all of those markets where demand exceeds supply) *must be exactly equal to* the total of excess supply (in all those markets where supply exceeds demand). It tells us that if any good is in excess supply, then one or more other goods must be in excess demand, and that overall the excess demand and excess supply will have identical magnitudes. It must be kept in mind, however, that this law refers not only to goods currently produced but also to money, securities, and, in general, all previously existing assets that could be exchanged.

We can see now why one of Walras's equations could be dropped in our discussion of his system of equations. If one market is out of equilibrium, then one or more other markets must simultaneously be out of equilibrium for excess demand to equal excess supply. Therefore, if all markets but one are in equilibrium, then the last market must also be in equilibrium. Therefore, the equations giving equilibrium conditions in all markets contain one unnecessary equation. This allowed Walras to arrive at precisely the right number of equations to solve for the unknown price and quantity variables. In the Appendix, we use a simple three commodity example to demonstrate some of the technical points of general equilibrium. Even with only three commodities the analysis becomes quite complex and a bit challenging to fully grasp. The main point, however, is the demonstration of the possibility for simultaneously explaining the equilibrium price and quantity for any number of commodities given Walras' assumptions concerning perfect competition and utility schedules.

Stability of General Equilibrium

A central issue in general-equilibrium theory is whether market forces will *automatically* correct a disequilibrium, that is, whether when a disequilibrium set of prices actually exists, the market forces of supply and demand will auto-

matically change these prices until equilibrium is established. If one believes this to be true, then there remains the very significant issue of how long this will take and what sorts of human costs or human suffering will be involved.

At stake in these issues are the different policy orientations of thinkers such as Bentham (in his later writings), Malthus, Marx, Hobson, and Keynes—all of whom argued that reliance on the market would involve enormous human costs and that, therefore, steps ought to be taken to mitigate these costs (although they advocated very different remedies)—and Say, Ricardo, Senior, Bastiat, and nearly all neoclassical economists—all of whom had a faith that such market adjustments would be fast and effective and should be relied on completely.

Suppose there exists an excess supply of all consumer commodities—in other words, a glut exists—matched by an excess demand for money. What will result in this situation? The answer to this question has divided economists from the time of Malthus and Say to the present. Walras stood squarely on the side of Say and all other proponents of extreme laissez-faire in maintaining that the market would automatically and relatively without cost create the appropriate full-employment equilibrium prices.

Walras described the process of the market attaining an equilibrium as a process of *tâtonnement* or "groping." In most of his discussions of this problem, he assumed that the economy *started* at a position of equilibrium, which was disturbed by a change in the demand for one commodity.[65] He always assumed that a situation of excess demand would immediately lead to a price increase and a situation of excess supply would immediately lead to a price decrease. These price changes would automatically reestablish equilibrium, either through altering the quantities people wished to exchange or through a reallocation of resources as enterprises rapidly converted from the excess supply industries to the excess demand industries: "The upward and downward movements of prices in conjunction with the effective flow of entrepreneurs from enterprises showing a loss to enterprises showing a profit is purely and simply a method of groping towards a solution of the equations involved in these problems."[66]

There were several problems with Walras's solution (which have never been satisfactorily resolved to the present time). First, for the ideological conclusions of his theory, as we shall see, Walras required the assumption of perfect competition. But in the neoclassical vision of perfect competition every firm is a *price taker.* Prices are first established in the market and then firms respond to those prices. How then do the new prices get established? Neoclassical economists have always had a most difficult time with this question.

To deal with this problem, Walras had to assume an auctioneer or "crier" to announce all prices to everyone:

> The markets which are best organized from the competitive stand point are those in which purchases and sales are made by auction, through the instrumentality of . . . criers acting as agents who centralize transactions in such a way that the terms of every exchange are openly announced and an opportunity is given to sellers to lower their prices and to buyers to raise their bids.[67]

But Walras's imaginary crier was not enough. Suppose the crier announced a set of prices and exchange took place. If this set of prices was *not* the equilibrium set, then excess demand and supply would result in many markets. Then the next set of prices proclaimed by our crier would find all individuals trying to achieve an equilibrium in terms of these new prices but also trying to rectify the mistakes made on the basis of the wrong prices previously announced. Unless by pure luck, the new set of prices will also be a disequilibrium set. All new exchanges and corrections of past mistakes result in a new round of mistakes. All traders compound mistake upon mistake, and the situation seems at least as likely to move away from equilibrium as toward it.

Walras had only two choices: either his crier could be omniscient (that is, be God) and know in advance what the equilibrium set of prices would be, or his crier would have to be the equivalent of a socialist central planning agency with many high-speed computers at its disposal. In this latter case, the crier could announce the price set and then every exchanger could respond by telling the crier *their intentions* to buy and sell at those prices. Then no one would act until the central planning agency fed all of this data into its computers and computed the total demand and supply for every market. When the crier with the aid of the central planning agency found some markets to have potential excess demands, the crier would adjust those prices upward. Similarly, markets with potential excess supply would have their prices adjusted downward. This process would continue as a series of gropings in which exchangers held their commodities until the crier discovered the equilibrium set of prices. Only then could they exchange.

Despite his aversion to socialism, Walras chose this crier and central planning agency model as a means of avoiding the issue of the "anarchy of the market" (to use the socialist phrase):

> Once the *prices* or the ratios of exchange of all these goods and services have been cried at random . . . each party to the exchange will *offer* at these prices those goods or services of which he thinks he has relatively too much, and will demand those articles or services of which he thinks he has relatively too little. . . . The quantities of each thing effectively demanded and offered having been determined in this way, the prices of those things for which the demand exceeds the offer will *rise,* and the prices of those things of which the offer exceeds the demand will *fall.* New prices having now been cried, each party to the exchange will offer and demand new quantities. And again prices will rise or fall until the demand and the offer of each good and each service are equal. Then the prices will be *current equilibrium prices* and [only then] exchange will effectively take place.[68]

Walras realized, of course, that such a crier did not exist. But he had a faith—and it was never anything more than faith—that the actual working of the market would be similar to this. One problem (but by no means the worst problem) for Walras's faith was that whenever any one price changed, the change affected not just the supply and demand in that one market, but those in numerous other markets as well. Thus, suppose all markets but two

were in equilibrium. In those two, one had excess demand and one had an equivalent excess supply. By Walras's reasoning, the price would rise in the excess demand market and fall in the excess supply market. If these were the only markets affected, and if supply and demand curves had the shapes that Walras had theorized, and if in fact the only response to the disequilibriums was that predicted by Walras, then the market would indeed establish a general equilibrium.

But these ifs must be rejected. First, the whole point of Walras's general equilibrium analysis was to show that price changes in any one market affected supply and demand in innumerable other markets. Thus, as the two prices began to change, many other supply and demand curves would shift. The initial disequilibrium would spread to many markets. As these other prices began to change, nearly all supply and demand curves would begin to shift again. Thus, disequilibrium in two markets could rapidly become a disequilibrium in all markets. How then would equilibrium ever be established?

Walras escaped this dilemma by asserting that a given price change would have a *primary effect* only in the market of the affected commodity. Its effects on other markets would be *secondary.* He then asserted his faith that "these secondary effects, however, will be less appreciable than the primary effect if there are a great many commodities . . . on the market."[69]

Subsequent disciples of Walras have shown that with all of Walras's assumptions, if the secondary effects are sufficiently small, then the market will indeed automatically achieve an equilibrium. But they have never shown that these secondary effects must, in reality, be that small.

Furthermore, several of Walras's other assumptions are also highly questionable. He assumed that because the economy was perfectly competitive, the small, relatively powerless business firms would always react to a situation of excess supply by lowering their prices. Experience has taught us, however, that large, powerful business firms that have some control over prices tend to reduce their level of output and attempt to maintain their price in the face of an excess supply that they view as temporary. This reduction in output reduces incomes, which further reduces demand for other products. If these producers react to the resultant excess supply in their market by reducing output, then a general glut, economic crisis, or depression would seem to be the only possible result.

Moreover, Walras assumed that all exchangers would react to any set of prices as if they expected those prices to be the equilibrium prices that would continue to prevail. Again, experience shows that exchangers frequently behave otherwise, exchanging on the basis of prices that they expect to obtain in the future. John Maynard Keynes showed, as we will see in a later chapter, that these expectations can frequently make the achievement of a full-employment general equilibrium impossible.

We can conclude that Walras's belief that the market would automatically create a full-employment general equilibrium was as much a matter of pure

faith as had been Say's belief—and both beliefs were equally unjustifiable on either theoretical or empirical grounds.

Walras's theoretical framework for his general equilibrium model was and still remains significant. If we drop his highly unrealistic faith in the automaticity of the market, his system of market interrelationships shows just how difficult it would be for a capitalist market system ever to achieve a full-employment general equilibrium. The theory can also show how, once a crisis starts, it spreads to all sectors of the economy and becomes a general crisis or depression. Walras's general-equilibrium framework is the best theoretical context within which to analyze the anarchy of the market. Many underconsumption theorists would have avoided innumerable logical and theoretical inadequacies had they formulated their theories within some framework akin to Walras's general-equilibrium theory. Therefore, Walras's theory of general equilibrium must be judged as one of the most significant theoretical achievements in the history of economic ideas. The theory can be easily extricated both from Walras's naive faith in the automaticity of the market and from his conservative, utilitarian ideology with which he justified competitive, laissez-faire capitalism.

Walras's Ideological Defense of Capitalism

Walras shared several characteristics with nearly all the economic theorists writing in the utility tradition, from Say and Senior to the present. First, he viewed the economy almost entirely from the exchange perspective. Even when he wrote about production, like most utility theorists, he viewed production primarily as a series of *exchanges* and never from the perspective of the class relationships involved in production. Thus, his analysis remained in what Marx called the sphere of circulation.

Walras's denial of the class relationships in the production process can be seen most clearly in his discussion of the entrepreneur. There were three types of productive services—capital, land, and labor. The pattern of ownership of these services was taken for granted. Each and every individual, acting identically as maximizers of utility, sold some quantity of productive services to an entrepreneur in order to buy some consumer goods—the whole process being merely an exercise in utility maximization through exchange. Each productive service was paid by the entrepreneur exactly the value of what it contributed to production. Interest was the payment to capital, rent to land, and wages to labor. In equilibrium, these productive payments exactly exhausted the value of what was produced; *there was no surplus value and there was no profit;* profit was only present when disequilibrium existed (and was matched by an equal loss somewhere else in the system).

In this view, who chose to be the entrepreneur was purely accidental. It might be a capitalist, who would then pay for labor services and land services and have a remaining residual (in equilibrium) exactly equal to the interest on

the services of his or her capital. It might just as well be a laborer, who would then pay for capital services and land services and have a remaining residual exactly equal to the wages for his or her labor services. Similarly, it could be a landlord or a person owning some combination of productive services who acted as an entrepreneur. Because profits were zero in equilibrium, it did not matter who was the entrepreneur: "Thus, in a state of equilibrium in production, entrepreneurs make neither profit nor loss. They make their living not as entrepreneurs, but as land-owners, labourers or capitalists in their own or other businesses."[70]

Therefore, all individuals were simply utility-maximizing exchangers with various initial "endowments" of exchangeable things. There was no difference between a Henry Ford and the poorest, lowest-paid worker on his assembly line. Both were maximizing exchangers, and production was merely a species of exchange. Such is the essence of Walras's theory, which has numerous adherents in contemporary academic economics.

Bastiat had declared that "Political Economy *is* exchange." After developing a much more elaborate and sophisticated theory than Bastiat, Walras came to an identical conclusion:

> Thus: The exchange of two commodities for each other in a perfectly competitive market is an operation by which all holders of either one, or both, of the two commodities can obtain the greatest possible satisfaction of their wants consistent with the condition that the two commodities are bought and sold at one and the same rate of exchange throughout the market.
>
> The main object of the theory of social wealth is to generalize this proposition by showing, first, that it applies to the exchange of several commodities for one another as well as to the exchange of two commodities for each other, and secondly, that under perfect competition, it applies to production as well as to exchange. The main object of the theory of production of social wealth is to show how the principle of organization of agriculture, industry and commerce can be deduced as a logical consequence of the above proposition. *We may say, therefore, that this proposition embraces the whole of pure and applied economics.*[71]

He made similar statements throughout his book. At one point, for example, he asserted that "everyone competent in the field knows that the theory of exchange . . . constitutes the very foundation of the whole edifice of economics."[72]

Like Senior, Bastiat, and nearly all neoclassical economists, Walras thought he saw a clear distinction between moral values and science. His writings were, he assured the reader, pure science, not defiled or vitiated by any value judgments. In the first chapter of the *Elements* he proclaimed: "Indeed the distinguishing characteristic of a science is the complete indifference to consequences, good or bad, with which it carries on the pursuit of pure truth."[73] Like Senior, Bastiat, and nearly all neoclassical economists, Walras honored this proclamation only in the breach. On the next page, he began immediately to show where "pure truth" stood in relation to controversial social issues: "It has been observed . . . that in times gone by, industry languished and stagnated under a system of guilds, trade regulations and price fixing. It is evident today

that under the opposite system of freedom of enterprise and freedom of trade, industry grows and prospers."[74]

The principal problem was that socialists had failed to understand the beneficence, prosperity, and harmony of capitalism. While prior economists who had advocated and defended laissez-faire capitalism had performed a valuable service to society by countering the claims of socialists, they had unfortunately formulated the correct defense of capitalism inadequately in Walras's opinion. For example, in his criticism of the Physiocrats, he concluded: "Mingled with their errors are found views of extraordinary profundity and accuracy; . . . [one] of enduring value . . . declares that for the production of wealth free competition is the best rule, subject to exceptions only when they can be justified."[75]

In discussing the superiority of his own theory of interest over those of the "English School" (by which he meant Ricardo and Mill), Walras wrote: "The theory of interest . . . has been a favourite target for socialists; and the answer which economists have given to these attacks has not, up to the present, been overwhelmingly convincing."[76] Similarly, while sympathetically criticizing J.B. Say and other economists who had taken Say's view, he wrote: "Such a point of view was particularly useful to them in their controversy with the socialists. . . . Unfortunately, convenient as this point of view is, it is mistaken."[77]

Walras's conception of his role in combating the ideas of socialists was again expressed in the following quotation:

> The equations we have developed do show freedom of production [Walras's phrase meaning capitalism] to be the superior general rule. Freedom procures, within certain limits, the maximum of utility. . . . Unfortunately, it must be said that up to the present economists have been less concerned with establishing proofs for their arguments in favour *of laisser-faire, laisser-passer* than they have been with using them as weapons against the socialists.[78]

Finally, Walras left no question as to the use to which his "pursuit of pure truth" should be put. In discussing communism and capitalism he wrote:

> It is a question of the relation of ethics to economics which was hotly debated by Proudhon and Bastiat, among others, around 1848. In his *Contradictions Economiques,* Proudhon argued that there is a conflict between justice and material well-being [Walras's euphemism for capitalism]. Bastiat in his *Harmonies Economiques* defended the opposite thesis. I think that neither proved his point. I shall take up Bastiat's proposition again and defend it in a different way.[79]

Much of Walras's *Elements* consisted of an elaborate defense of Bastiat's thesis that in the existing capitalist society, free exchange maximized total utility, and, hence, laissez-faire capitalism was the best of all possible worlds.

Beginning with his assumption that "the ownership of property . . . is . . . in conformity with justice,"[80] Walras went on to show that all problems in economics could be reduced to problems of exchange: "Exchange of several

commodities for one another in a market ruled by free competition is an operation by which all holders of one, several or all of the commodities exchanged can obtain the greatest possible satisfaction of their wants."[81]

This statement appeared in various forms at least twenty times in Walras's *Elements*. It is clear that he believed that competitive capitalism inevitably maximized social welfare and that this was a mere scientific deduction from postulates that represented "pure truth," uncontaminated by value judgments. Ironically, certain isolated passages in Walras's own writings contain the germ of a devastating moral critique of his own utilitarianism. Bentham's statement that "quantity of pleasure being equal, pushpin is as good as poetry" finds its equivalent in the following statement by Walras:

> From other points of view the question of whether a drug is wanted by a doctor to cure a patient, or by a murderer to kill his family is a very serious matter, but from our point of view, it is totally irrelevant. So far as we are concerned, the drug is useful in both cases, and may even be more so in the latter case than in the former.[82]

Thus, in the utility perspective, total social satisfaction may be maximized when all of the drug goes to the murderer to kill his family and none goes to the doctor to cure his patients. Walras could have shown with the same Benthamite logic that milk being used for a rich person's beauty bath contributes more to the social welfare than milk for a starving baby of poor parents.

Intellectual Perspective of Neoclassical Marginalism

At the beginning of this chapter, we described the process of the concentration and centralization of industry that was occurring during the period in which neoclassical marginalism received its first significant formulations (the early 1870s). During the last three decades of the nineteenth century, the concentration of industry grew at an accelerated pace and neoclassical marginalism (as first formulated by Jevons, Menger, and Walras) came to dominate completely the orthodox, conservative tradition in economic theory. The concurrence of this change in the socioeconomic structure of capitalism and the emergence of marginalism in economic theory do not seem to be totally unrelated.

During the initial phases of the industrial revolution (from the middle of the eighteenth century through the first few decades of the nineteenth century), industrial capitalists engaged in a prolonged struggle against the landed interests and merchant capitalists for economic and political supremacy. During this period, industrial capitalists usually were personally involved in directing, coordinating, and overseeing the actual processes of production. The central focus or objective of their endeavors had been the rapid accumulation of industrial capital, and their main intellectual concern had been to understand the source of capital accumulation. The labor theory of value perspective had furnished the most serviceable insights into the process of capital accumulation, focusing on the distinction between productive and unproductive labor.

It had shown how productive labor was the source of the surplus labor that made the expansion of capital possible. Thus, in its earliest formulations, the labor theory of value reflected the perspective of, and was serviceable in the fulfillment of the objective needs of, the industrial capitalist class.

During this same period, merchant capitalists and landlords received their incomes from ownership and market exchange. Their situation could best be served by economic theories that sanctioned private ownership of capital and land, while extolling the social beneficence of exchange. The exchange or utility perspective, as represented in the writings of Malthus, Say, and Senior, had serviced their needs best.

With the growth of the corporation as the principal form of industrialization and the growing industrial concentration described at the beginning of this chapter, there was an important change in both the nature of the accumulation of industrial capital and the role of the industrial capitalist. The accumulation became systematized, institutionalized, and regularized. Increasingly, corporate managers were hired to direct and oversee industrial enterprises and to channel profits automatically as part of a perpetual accumulation process. The role of the individual capitalist entrepreneur in the industrial production processes became increasingly less significant.

The owners of industrial capital came more and more to resemble, in social and economic functions, the landlord class. Increasingly, profits and interest came to be the result of passive ownership. Therefore, the theoretical and ideological needs of the owners of industrial capital became identical with those of the landlords and merchant capitalists. They all needed a theory that sanctioned their ownership and proclaimed the virtues of an exchange economy.

Therefore, at the very time when the labor theory of value, in the writings of Karl Marx, was becoming identified with the interests of the working class, the utility theory or market perspective began to serve the interests of all elements of the class owning the means of production (whether land, merchant capital, finance capital, or industrial capital).

The individual as a rational, calculating maximizer, as portrayed in neoclassical marginalism, has never been an accurate reflection of the behavior of most people in a capitalist society. Most working-class people have grown up in families where their consumption patterns were socialized so as to become habitual and relatively standardized. To be sure, if a commodity that they habitually consumed rose sharply in price, their limited purchasing power would force them to adjust their consumption patterns. But the thought of rationally calculating marginal utilities, comparing utility ratios to price ratios, and adjusting their purchases so as to attain a "maximum of pleasure" has always been utterly foreign to the mental processes of most working-class people.

This view of the individual as a rational, calculating maximizer who buys at this margin and sells at that margin in a constant quest to maximize is, however, quite descriptive of one group in a capitalist economy: the functionless owner of a broad portfolio of investment assets. Such an individual owns a variety

of stocks, bonds, land, and other assets yielding income to ownership alone. Such an individual, usually working through specialized exchange brokers, is constantly selling some of this stock, buying some of that stock, shifting from short-term bonds to long-term bonds, or making perpetual *marginal adjustments* to his or her portfolio of assets in an unceasing effort to calculate rationally the combination of property holdings that will maximize either income through time or the rate of growth in the value of owned assets. This is the type of individual best described by neoclassical marginalism, and neoclassical marginalism culminates in ideological conclusions that best service the needs of this type of individual.

Appendix

It is very difficult to grasp intellectually the real meaning of a general equilibrium from a discussion of equations. It is even more difficult if one merely looks at $2m + 2n–1$ actual equations (particularly if this represents a very large number). Therefore, we will graphically present a simple three-commodity general-equilibrium model. This will help to illustrate the nature of the general-equilibrium model and serve to indicate the problem of how equilibrium may or may not be attained when the economy starts from a position of disequilibrium. Our model will have no distinction between productive services and consumer goods. It is simply a model of exchange—not production—in which one commodity serves as a *numeraire*. There are five unknowns—the quantities exchanged of commodities *a, b,* and *c,* as well as the price of $a(P_b)$ and the price of $b(P_b)$, both stated in terms of the numeraire, commodity *c*.

Our equations are:

$$ED_a = f(P_a, P_b), \tag{10A.1}$$

that is, the excess demand (ED) for a depends on both prices. This simply means that for any set of the two prices, one of three situations will arise in the market for *a:* $ED_a = 0$, that is, the supply of and demand for a will be equal; $ED_a > 0$, that is, the demand for a will exceed the supply; or $ED_a < 0$, that is, the demand for a will be less than the supply of *a;* negative excess demand is identical to excess supply.

$$ED_b = f(P_a, P_b), \tag{10A.2}$$

that is, the excess demand for b depends on both prices; the meaning of this function that was given for commodity a applies equally to commodities b and c.

$$ED_c = f(P_a, P_b), \tag{10A.3}$$

that is, excess demand for c also depends on both prices.

$$ED_a = 0 \tag{10A.4}$$

that is, excess demand for a must be zero, as a condition of equilibrium.

$$ED_b = 0 \tag{10A.5}$$

the same equilibrium condition must apply for commodity b.

We now have five unknowns and five equations. Walras's law assures us that if a and b are in equilibrium, c will also be in equilibrium. We could just as well have had eight unknowns and eight equations—the unknowns would have been the two prices and the individual supply and demand functions for each of the three commodities. Actually, Equations 10A.1, 10A.2, and 10A.3 are each really the combination of a supply equation and a demand equation. They are combined because in equilibrium the quantity of the supply of any good equals its demand, and both equal the quantity exchanged.

We can now illustrate general equilibrium with the use of simple supply and demand curves taken from partial equilibrium analysis. We do it in the following way: in the market for a, we can *assume* a price for b and then construct supply and demand curves showing how much of a would be supplied and demanded at different prices of a, given the assumed price of b. We next assume a *different* price for b, which gives us two entirely different supply and demand curves for a, under the same assumption. If we were to continue to vary the price of b, we get a family of supply and demand curves for a. These curves, together with the different prices of b that generate them, give us the information implicit in Equation 10A.1. For any price of b, there will be one supply curve and one demand curve for a. At some particular price of a, those curves will cross; at that price, Equation 10A.4 is solved, *given the assumed price of b*. The assumed price of b and the corresponding price of a that satisfies Equation 10A.4, that is, which equates supply of and demand for a, will give us *one set* of prices that satisfies Equation 10A.4. Each time we vary the price of b, we will get another set of prices that satisfy Equation 10A.4.

Figure 10A.1 illustrates the foregoing explanation. The first supply and demand curves (solid lines) are constructed with a given price of b, P_{b1}, and the second set of curves (broken lines) are constructed with a second given price P_{b2}.

In part (a) of Figure 10A.1, we see that with the supply and demand curves for a corresponding to P_{b1}, the equilibrium of supply and demand is at P_{a1}. Similarly, with those corresponding to P_{b2}, the equilibrium is at P_{a2}. Thus, we have two sets of prices for a and b that satisfy Equation 10A.4. If we continued this process, we could get a whole series of price sets that satisfy Equation 10A.4. Part (b) of Figure 10A.1 is a graph depicting *all* sets of prices satisfying Equation 10A.4.

The line labeled $ED_a = 0$ is the locus of all points satisfying Equation 10A.4. The graph in part (b) shows prices of a (on the vertical axis) and prices of b

(on the horizontal axis). Any point on the line $ED_a = 0$ gives two prices that will equilibrate the supply and demand for *a*. Point 1 on that line corresponds to the first set of supply and demand curves (in solid lines) in part (a), and point 2 corresponds to the second set of supply and demand curves (in broken lines) in part (a).

One further point should be made about part (b) of Figure 10A.1. All points in the space above and to the right of the $ED_a = 0$ line correspond to sets of prices for *a* and *b* that will result in a negative excess demand (or an excess supply) for *a*. All points below and to the left of the line are price sets that will result in excess demand (or a negative excess supply) for *a*.

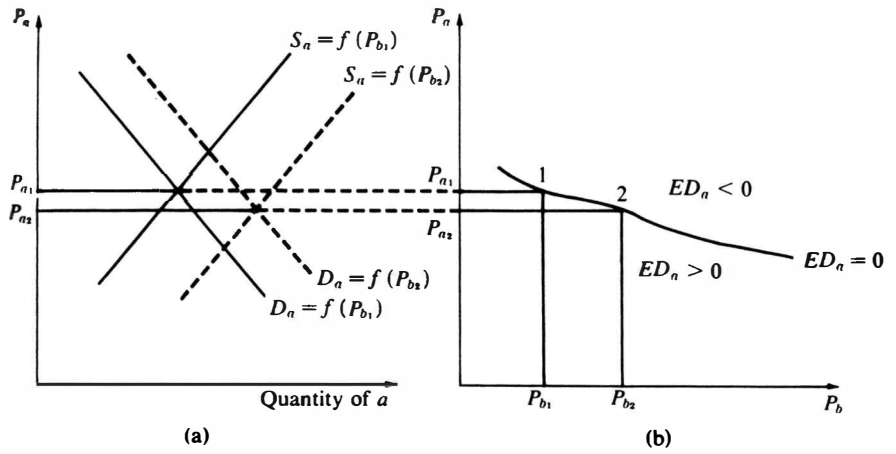
Figure 10A.2 illustrates exactly the same things for commodity *b*. Parts (a) and (b) of the graph have exactly the same rationale as those of Figure 10A.1. In Figure 10A.3 we have combined the part (b) graphs from both Figures 10A.1 and 10A.2. Here we see in the same graph all sets of prices that will equilibrate the supply and demand for both commodities. We see that only $\overline{P}_a$ and $\overline{P}_b$ will result in an *equilibrium in both markets*. All points not on the $ED = 0$ lines are divided into four regions. The excess demand conditions for both *a* and *b* at every point in each of these four regions are summarized to the right of the graph in Figure 10A.3.

On the basis of the information illustrated in Figure 10A.3 and Walras's law, we can construct another line giving all sets of prices for which the excess demand for $c(ED_c)$ will equal zero. First, from Walras's law, we know that the point $(\overline{P}_a, \overline{P}_b)$ at which the $ED = 0$ lines of *a* and *b* intersect must be a point where $ED_c = 0$. We did not need an equation (in our original five equations of our general equilibrium system) for the equilibrium of commodity *c,* because we knew that if all other commodities were in equilibrium, then commodity *c* had to be in equilibrium. For *c* to be in disequilibrium would require that at least one of the other two commodities be in disequilibrium, because total demand must equal total supply by Walras's law. Therefore, point $(\overline{P}_a, \overline{P}_b)$ is a point on the $ED_c = 0$ line.

But there must be other sets of prices where $ED_c = 0$, and the locus of all points representing these sets of prices will give us the $ED_c = 0$ line. We know that at any set of prices other than $(\overline{P}_a, \overline{P}_b)$, either the *a* market or the *b* market or both will be in a state of disequilibrium (because $\overline{P}_a$ and $\overline{P}_b$ constitute the only set of prices that will equilibrate *both* the *a* market and the *b* market). Therefore, all sets of prices on the $ED_c = 0$ line other than $(\overline{P}_a, \overline{P}_b)$ must involve an equilibrium in the *c* market and a disequilibrium in both the *a* and *b* markets (because from Walras's law we know that it is impossible for only one market to be in disequilibrium).

Moreover, from Walras's law, we know that at all points on the $ED_c = 0$ line other than $(\overline{P}_a, \overline{P}_b)$ the disequilibriums in the *a* and *b* markets must offset each other exactly in order that total demand remain equal to total supply. Therefore, at any such point on the $ED_c = 0$ line, if $ED_a > 0$, then it must be the case that $ED_b < 0$, and the magnitude of the excess demand for *a* must equal

Figure 10A.1 **Prices Equilibrating the Supply of and Demand for Commodity *a***



the magnitude of the negative excess demand (excess supply) for *b*. Similarly, if $ED_a < 0$, then it must be the case that $ED_b > 0$, and the magnitudes of the excess demand and excess supply must again be equal.

Looking at Figure 10A.3, we see that the $ED_c = 0$ line cannot pass through regions *F* or *H*, because in these regions the markets for *a* and *b* both have either excess supply (region *F*) or excess demand (region *H*). Only in regions *J* and *G* is it possible for the *c* market to be in equilibrium and for the disequilibriums in the *a* and *b* markets to offset each other.

Therefore, if we find every set of prices in region *J* where the excess supply of *a* exactly equals the excess demand for *b*, and every set of prices in region *G* where the excess demand for *a* exactly equals the excess supply of *b*, the resulting locus of points will constitute the $Ed_c = 0$ line.

Figure 10A.4 illustrates the lines giving the sets of equilibrium prices for each of the three commodities. The prices $\overline{P}_a$ and $\overline{P}_b$ equilibrate all three markets. They are therefore the general equilibrium prices. Two of our original five unknowns are now solved and known. With these two prices we can go back to our supply and demand curves for each commodity (as illustrated in part (a) of both Figures 10A.1 and 10A.2) and see exactly how much of each commodity is exchanged. We did not illustrate the supply and demand curves for commodity *c*. They would be exactly the same as those of *a* and *b* with one difference: because the price of *c* is defined as the *numeraire* (and is therefore always equal to one), we could either make the demand and supply schedules for *c* depend on the price of *a* with the price of *b* given (and then vary the price of *b* to generate new supply and demand schedules), or make the supply and demand schedules for *c* depend on the price of *b* with the price of *a* given (and then vary the price of *a* to generate new supply and demand schedules). Both methods would yield identical results.

Figure 10A.2 **Prices Equilibrating the Supply of and Demand for Commodity *b***

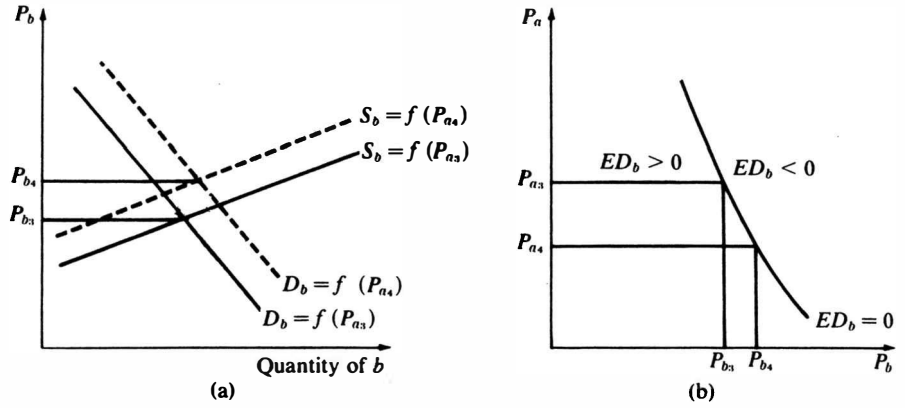
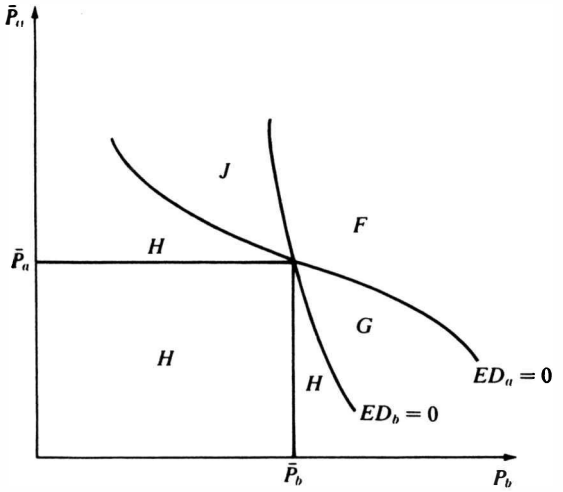


Figure 10A.3 **Equilibrium in the *a* and *b* Markets and the Four Regions of Disequilibrium**



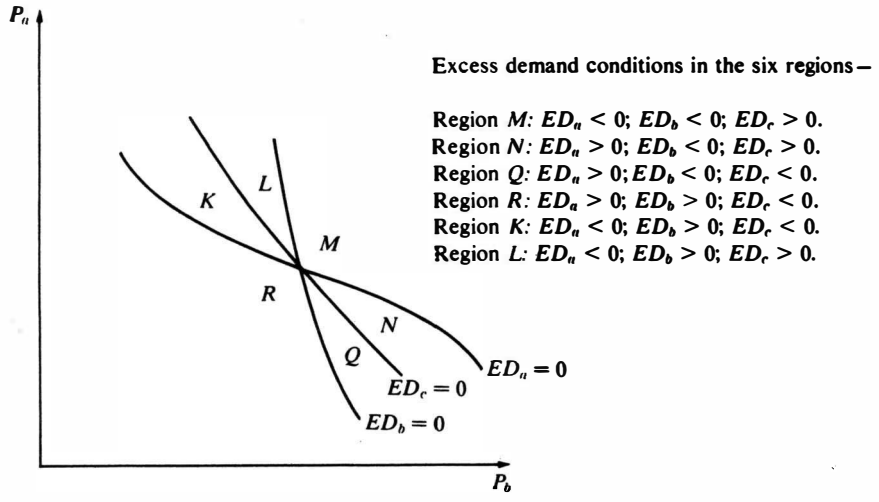
Excess demand conditions in the four regions –

Region *F*: $ED_a < 0$; $ED_b < 0$.
Region *G*: $ED_a > 0$; $ED_b < 0$.
Region *H*: $ED_a > 0$; $ED_b > 0$.
Region *J*: $ED_a < 0$; $ED_b > 0$.

Therefore, starting with five unknowns and five equations, we have seen how to determine the general equilibrium solutions for both prices and the quantities exchanged of each of the three commodities. We can appreciate the complexity of a general equilibrium theory explaining the prices and quantities exchanged of a large number of both consumer goods and productive services by understanding the complexity of our simple three-commodity general-equilibrium model.

Figure 10A.4 illustrates all three $ED = 0$ lines. We now divide the space between these three lines into six regions. At the right of the graph, the excess demand and supply conditions in each of these regions are summarized. Figure 10A.4 can be used to graphically demonstrate the issue of whether market

Figure 10A.4 **Equilibrium in the *a*, *b*, and *c* Markets and the Six Regions of Disequilibrium**



forces will automatically correct a disequilibrium. Suppose, for example, prices begin in region M and commodity c is money. Commodities a and b would be in excess supply (or, negative excess demand) while commodity c was in excess demand. Hence, an excess supply—or general glut—exists for all commodities, matched by an excess demand for money. Prices for a and b should both fall in terms of the *numeraire* commodity c to move the markets toward the equilibrium solution—graphically, the intersection of the three lines. The issue then is whether this movement towards the equilibrium will in fact occur, and if so just how long will it take.

Notes to Chapter 10

1. William Stanley Jevons, *The Theory of Political Economy,* 2d ed., ed. R.D. Collison Black (Baltimore: Penguin, 1970).
2. Carl Menger, *Principles of Economics* (New York: Free Press, 1950).
3. Leon Walras, *Elements of Pure Economics* (Homewood, IL: Irwin, 1954).
4. William Stanley Jevons, *Letters and Journal of W. Stanley Jevons,* ed. Harriet A. Jevons (London: Macmillan, 1886), p. 151.
5. Jevons, *Theory of Political Economy,* p. 55.
6. Ibid., p. 44.
7. Ibid., p. 77.
8. Ibid., p. 127.
9. Ibid., pp. 101–2.
10. Ibid., p. 101.
11. Ibid., p. 111.
12. Jevons used different symbols than we have used. The meanings of his formulas were, however, identical to those used here (see Ibid., pp. 142–43). We have used the standard contemporary form of the equation to avoid confusing the reader.
13. Ibid., p. 44.
14. Ibid., p. 52.
15. Ibid., p. 85.

16. William Stanley Jevons, *The State in Relation to Labour* (London: Macmillan, 1882), p. 98.
17. Ibid., p. 104.
18. Jevons, *Theory of Political Economy,* pp. 68–69.
19. Ibid., p. 173.
20. Ibid., p. 256.
21. Ibid., pp. 257–58.
22. Jevons, *State in Relation to Labour,* p. 127.
23. William Stanley Jevons, *Methods of Social Reform and Other Papers* (London: Macmillan, 1883), p. 108.
24. Jevons, *Theory of Political Economy,* p. 126.
25. Ibid., p. 257.
26. Ibid., p. 225.
27. Ibid., p. 72.
28. Ibid., p. 101.
29. Ibid., p. 103.
30. Jevons, *Methods of Social Reform,* p. 196.
31. Quoted in John Maynard Keynes, *Essays in Biography* (London: Macmillan, 1933), pp. 126–27.
32. Carl Menger, *Problems of Economics and Sociology* (Urbana: University of Illinois Press, 1963).
33. Menger, *Principles of Economics,* p. 127.
34. Ibid.
35. Ibid., ch. 5.
36. Ibid., pp. 222–25.
37. Ibid., pp. 80–87.
38. Ibid., p. 85.
39. Ibid., p. 150.
40. Ibid., p. 149.
41. Ibid., p. 156.
42. Ibid., pp. 164–65.
43. Ibid., p. 161.
44. Ibid., p. 75.
45. Ibid., p. 76.
46. Ibid., p. 97.
47. Ibid., pp. 173–74.
48. Menger, *Problems of Economics and Sociology,* p. 237.
49. Ibid., p. 194.
50. Ibid., p. 195.
51. Ibid., p. 194.
52. Ibid., p. 91.
53. Ibid., p. 230.
54. Ibid., pp. 91–92.
55. Menger, *Principles of Economics,* p. 174.
56. Ibid., p. 174.
57. Walras, *Elements of Pure Economics,* p. 42.
58. Ibid., p. 67.
59. Ibid., pp. 75, 80.
60. Ibid., p. 78.
61. Ibid., pp. 115–17.
62. Ibid., p. 117.
63. Ibid., p. 69.
64. Ibid., p. 89.
65. Ibid., pp. 146–47, 178–79.
66. Ibid., p. 44.
67. Ibid., pp. 83–84.
68. Ibid., p. 179.

69. Ibid.
70. Ibid., p. 225.
71. Ibid., p. 143; italics added in the last sentence.
72. Ibid., p. 44.
73. Ibid., p. 52.
74. Ibid., p. 53.
75. Ibid., pp. 396–97.
76. Ibid., p. 422.
77. Ibid., pp. 54–55.
78. Ibid., p. 256.
79. Ibid., pp. 79–80.
80. Ibid., p. 67.
81. Ibid., p. 173.
82. Ibid., p. 65.

Chapter 11

Neoclassical Theories of the Firm and Income Distribution: The Writings of Marshall, Clark, and Böhm-Bawerk

The utility perspective in economic theory was incomplete until the entire economic process, as envisioned and defined in this tradition, could be shown to be wholly the result of rational, calculating, maximizing behavior. The economic process is seen, in this perspective, as having two important focal points—the household and the business firm. There are two continuous circular flows between these points. First, there is what economists call a "real flow." Households are viewed as the owners of the "factors of production" and as the consumers of final consumption goods. Based on a calculation of their marginal utilities, they decide how much to sell to business firms of each of their various factors of production and how much to buy of each of the various consumption goods. The real flow therefore is a flow of the use of productive factors from households to business firms and a return flow of consumption goods from business firms to households.

The second flow is a "monetary flow." Households receive money income from the sale of the use of their productive factors. This money then flows back to the business firms in payment for the consumption goods purchased by the households. The business firms make rational, calculating, maximizing decisions in a manner strikingly similar to the decisions of the households.

Whereas the household attempts to maximize the excess of the utility derived from the consumption goods purchased over the utility given up in the sale of productive factors, the business firm attempts to maximize the difference between the money it pays for productive factors and the money it receives from the sale of consumption goods. Therefore, the household is the focal point of the real flow, and utility is the thing being rationally maximized. Likewise, the firm is the focal point of the monetary flow, and profit is the thing being rationally maximized. Because the mathematical logic of maximization is the

same regardless of the use to which it is put, the analyses of the household and the firm are highly similar in neoclassical economics.

Marshall's Contribution to Utility Theory and Demand Theory

Alfred Marshall (1842–1924) was a mathematician turned economist who taught economics at Cambridge University for several decades. Although he had developed and was teaching most of his ideas by the early 1870s, he did not publish a full version of them until 1890. His *Principles of Economics* gradually came to replace Mill's *Principles of Political Economy* as the dominant economics text used in English-speaking universities. Many of his formulations of neoclassical theory have continued to dominate the teaching of introductory neoclassical microeconomic theory in colleges and universities to the present.

In general, it may be said that Marshall was somewhat closer in spirit and ideology to John Stuart Mill than he was to Senior, Say, and Bastiat, although Marshall was decidedly more conservative than Mill. Marshall was the founder of that portion of the twentieth-century neoclassical tradition that tempers its advocacy of laissez-faire capitalism with a considerable latitude given to minor reforms designed to make the economic system function less harshly. This tradition continues to analyze imperfections in markets, such as imperfect information, with the intent of informing economic policy to achieve superior outcomes. Many of the so-called progressive mainstream economists of our day can be said to come out of this tradition.

Because we have already discussed utility theory in the previous chapter, we will limit our discussion of Marshall's contributions to utility theory to those areas in which he went beyond Jevons, Menger, and Walras. Like the latter three theorists, he formulated the notion of diminishing marginal utility as well as the necessary conditions for consumer utility maximization through exchange. Neither Jevons nor Menger succeeded in directly linking their utility theory to demand theory. While Walras had shown such a link, Marshall was much more successful than Walras in rigorously deriving the contemporary neoclassical conclusions of demand theory from his notion of diminishing marginal utility.

Marshall deduced a negative demand curve in the following manner: first, as with all utility theorists, he assumed that during the period being analyzed the individual's schedule of marginal utilities remained unchanged. "We do not suppose," he wrote, "time to be allowed for any alteration in the character or tastes of the man himself."[1] We have already pointed out the importance of this assumption for the ethical theory underlying neoclassical economics. It was also a necessary assumption for Marshall's derivation of a demand curve. Second, he assumed that an individual's marginal utility of money was given and constant throughout the period of analysis. By assuming a fixed marginal utility of money, Marshall found a connecting link between utility schedules and price schedules.

For example, if a person has a marginal utility of two "utils" for $1 of money, and one additional loaf of bread would yield four utils while one additional

pound of steak would yield six utils to that person, then he or she would be willing to give up $2 (i.e., four utils) for a loaf of bread and $3 (i.e., six utils) for a pound of steak. The utility of a second additional loaf of bread might fall to two utils and that of a second pound of steak to three. In that case, he or she would be willing to pay $1 (two utils) for the second additional loaf of bread and $1.50 (three utils) for the second additional pound of steak. This was because "the marginal utility of money to him is a fixed quantity, so that the prices he is just willing to pay for two commodities are to one another in the same ratio as the utility of those two commodities."[2]

From this Marshall derived an individual demand schedule. The total demand curve for any commodity was simply the summation of the individual demand curves. "Every fall, however slight in the price of a commodity in general use, will, other things being equal, increase the total sales of it."[3]

The phrase "other things being equal" was significant. Although Marshall briefly discussed the conditions necessary for a general equilibrium, most of his theories were *partial equilibrium* analyses, in which he examined only the markets for one or two commodities and ignored the mutual interconnections between these markets and the markets for all other commodities.

In addition to explaining the derivation of a demand schedule from a utility schedule, Marshall defined and elaborated the notion of the "price elasticity of demand,"[4] defined and discussed his notion of "consumer's surplus,"[5] and showed how exceptional circumstances could result in an upward sloping demand curve.[6] If we were concerned about elaborating the origins of every detail of modern neoclassical analysis, these notions would be explained. For our purposes this is unnecessary; the interested reader is advised to consult an appropriate text.[7]

Thus Marshall developed one-half of the utility-maximizing process of the household by extending and elaborating the ideas of Jevons, Menger, and Walras. The other half of the household-maximizing problem concerned the sale of factors of production. Of the three traditional factors of production—land, labor, and capital—Marshall believed that only the furnishing of labor and capital involved a negative utility, or disutility. His discussion of rent was basically similar to those of Ricardo and Mill. Rent was a surplus created by the differing fertilities of the soil and involved no social cost.[8] In this as in other cases, Marshall's defense of income from ownership was not as complete or as extreme as those of Malthus, Say, Senior, Bastiat, and most other neoclassical economists.

It was in selling the services of labor and capital that households had to make utility calculations. After stating certain qualifications to the principle, Marshall asserted that when labor was sold, there was always a point beyond which "the marginal disutility of labour generally increases with every increase in its amount."[9] This increasing disutility of labor might

> arise from bodily or mental fatigue, or from its being carried on in unhealthy surroundings, or with unwelcome associates, or from its occupying time that is wanted for recreation, or for social or intellectual pursuits. But whatever be the form of the discommodity, its intensity nearly always increases with the severity and the duration of the labour.[10]

While it was always clear to all economists that labor was a social cost of production, conservative economists from Senior onward had argued that the capitalists' abstinence involved a similar social cost and a similar disutility. Marshall generally agreed with Senior but evidently was a bit embarrassed by the implications of this notion:

> Karl Marx and his followers have found much amusement in contemplating the accumulations of wealth which result from the abstinence of Baron Rothschild, which they contrast with the extravagance of a labourer who feeds a family of seven on seven shillings a week; and who, living up to his full income, practices no economic abstinence at all.[11]

Marshall hoped he could avoid this embarrassment by substituting the word *waiting* for *abstinence:*

> Human nature being what it is, we are justified in speaking of the interest on capital as the reward of the sacrifice involved in the waiting for the enjoyment of material resources, because few people would save much without reward. . . .
>
> The sacrifice of present pleasure for the sake of the future, has been called *abstinence* by economists. . . . Since, however, the term is liable to be misunderstood, we may with advantage avoid its use, and say that the accumulation of wealth is generally the result of a postponement of enjoyment, or a *waiting* for it.[12]

Thus, in Marshall's analysis of the household, people were always calculating the appropriate exchanges involved in minimizing the pains of labor or waiting against the pleasures received from acquiring consumption commodities. In doing so, "they strive to adjust their parings down so that the aggregate loss of utility may be a minimum."[13] The same utility calculations were made by laborers deciding how much painful exertion to endure as by capitalists deciding how much painful waiting to endure. "A prudent person will endeavor to distribute his means between . . . present and future in such a way that they will have in each the same marginal utility."[14] Therefore, utility-maximizing calculations controlled the real flow of the services of capital and labor from the household sector to the business sector and of the consumption commodities from the business sector to the household sector. To understand the monetary flow, however, it was necessary to examine the profit-maximizing behavior of the business firm.

Symmetry Between Neoclassical Theories of the Household and the Firm

Utility maximization through marginal adjustments of quantities of commodities bought and sold was possible in neoclassical theory because of the *substitutability* of any commodity for other commodities. Commodities were purchased only because they produced utility in the consumer. In maximizing utility, the consumer could derive utility from any one of innumerable com-

modities. If the price of a given commodity increased, then the cost to the consumer of deriving utility from that commodity increased. Because utility was conceived of as qualitatively homogeneous (in accordance with Bentham's view but in opposition to Mill's), the only considerations of a consumer were the quantities of his marginal utilities produced by different commodities and the costs of these commodities. Therefore, when the cost of a commodity increased, the consumer *substituted* some amounts of *other commodities* for a part of his consumption of the more expensive commodity. In this manner, he reduced his purchases of the commodity from which the derivation of marginal utility was relatively more costly and increased his purchases of other commodities from which the derivation of utility was relatively less costly. Thus, after the price increase, these shifts in his sales and purchases would reestablish the conditions for utility maximization (i.e., $MU_a / P_a = MU_b / P_b$, and so forth for every commodity), because of the substitutability of commodities in producing utility.

In order for the theory of the business firm to be stated in terms of a maximization problem similar to the maximization problem of the household, it was necessary to see the factors of production as analogous to consumption goods and the revenues received from the sale of the output produced by these factors as analogous to utility. The household gave up utility in selling the services of the factors of production in order to pay the costs of consumption goods. The consumption goods produced utility. Households attempted to maximize the difference between the utility produced by their consumption of these goods and the utility lost in paying the costs necessary to acquire the goods.

The problem for the firm in Marshall's analysis was identical. A firm wanted to maximize the difference between the money revenue it received from selling commodities and the money costs it paid for acquiring the services of the productive factors that produced the commodities; that is, it wanted to maximize profit. The firm bought factors of production, which by producing sellable commodities, produced revenue for the firm. The maximization problem for the firm would be essentially identical to that of the household *if the factors of production were substitutable in the process of producing revenue, in the same way that consumption goods were substitutable in the production of utility.*

But most of the classical economists had assumed that production involved *fixed technical coefficients* of production. That is, they assumed that a given productive technology implied a "recipe" for producing commodities that dictated fixed proportions of the different productive factors to be used. They generally ignored the possibility of varying the proportion of productive factors by substituting one factor for another (although Ricardo had asserted that when inferior land was brought into cultivation, the more fertile land would be farmed more intensively through the use of relatively greater amounts of capital and labor on a given parcel of land).

Menger had also assumed fixed technical coefficients of production. This was why he had assumed that productive factors were complementary and

why he had a difficult time explaining how each individual productive factor was priced in the market. In the first two editions of Walras's *Elements,* he also assumed fixed technical coefficients of production. Only in the third edition of the *Elements* (published in 1896) did Walras introduce the possibility of varying the proportions of productive factors by substituting one factor for another in response to changes in their relative prices.

Therefore, one of Marshall's most important contributions to utilitarian economic theory was his introduction of two closely related concepts: first, businessmen tried to reduce the costs of production by substituting one factor (or "agent," as Marshall sometimes referred to factors) for another factor:

> Every businessman . . . is constantly endeavouring to obtain a notion of the relative efficiency of every agent of production that he employs; as well as of others that might possibly be substituted for some of them.[15]

> And the sum of the prices which he pays for those factors which he uses is, as a rule, less than the sum of the prices which he would have to pay for any other set of factors which could be substituted for them.[16]

Second, when the firm increased the quantity used of one factor (e.g., labor) relative to the quantity used of another factor (e.g., capital), the marginal increment added to total production by each additional equal increment of the first factor (labor) would, beyond some point, begin to diminish in magnitude: "The notion of the marginal employment of any agent of production implies a . . . tendency to diminishing return from its increased employment."[17]

The substitutability of productive factors for the firm was thus analogous to the substitutability of consumption goods for the household. Similarly, the law of diminishing returns from the increased use of a productive factor was analogous to the law of diminishing marginal utility from the increased consumption of a commodity. From Marshall onward, the theory of the maximizing firm has been almost identical analytically to the theory of the maximizing household.

The firm purchased inputs and sold outputs. It attempted to maximize the difference between the costs of its inputs and the revenues from its outputs. The firm's maximization problem, therefore, could be viewed from either of two vantage points: the firm's outputs or its inputs. From the former perspective, one would examine the price at which the firm could sell various quantities of output and then calculate the firm's *total revenue* from the sale of its output, its *average revenue* per unit of output sold, and its *marginal revenue* arising from a small increase or decrease in the number of units of output sold. Similarly, one would calculate the firm's *total costs* of producing its output, its *average cost* per unit of output produced, and its *marginal cost* arising from a small increase or decrease in the number of units of output produced. From the perspective of a firm's inputs, one would examine the revenue resulting from the sale of the total produce of all of the units of a factor hired by the firm, the value of

the average output resulting from one unit of the factor, and the value of the increment or decrement to production caused by a small increase or decrease in the quantity of the factor hired by the firm. Similarly, one would examine the firm's total cost of purchasing the services of the factor in question, the average cost per unit of the factor purchased, and the marginal cost of a small increase or decrease in the amount of the factor purchased.

Looking at the firm's maximization problem from the vantage point of output is usually associated with the theory of the firm in contemporary economic literature. It was this aspect of Marshall's analysis that was to become the basis for most subsequent elementary expositions of neoclassical microeconomic theory. We will therefore examine Marshall's explanation of the firm's attempt to maximize profits from this vantage point.

Looking at the firm's maximization problem from the vantage point of inputs has become the basis of the neoclassical theory of the distribution of income. This theory—called the marginal productivity theory of distribution—was inadequately developed by Marshall. Superior formulations of the theory were developed independently by an Englishman, P.H. Wicksteed, and an American, John Bates Clark. After indicating the nature of the deficiency in Marshall's distribution theory, we will examine Clark's version of the theory. Finally we will examine the difference between Clark's conception of capital and that of Böhm-Bawerk (a difference that has remained important in neoclassical theory to the present time).

Marshall's Theory of the Firm

Marshall's analysis of the firm was an integral part of his analysis of price determination. Prices were determined by supply and demand. Demand was determined by consumers' utility schedules, and supply was determined by firms' costs schedules. Although Marshall briefly discussed monopolies on the sale of particular products, most of his *Principles* was devoted to analyzing the situation in which an industry consisted of numerous competing firms. With competition, the price of a commodity produced in an industry was determined by the total demand and supply of the entire industry. The firm typically took the price as given (with an exception that we will discuss shortly) and adjusted its output and costs so as to maximize profits.

Marshall's theory depicted what he called a "representative firm" in a competitive industry. The representative firm was "in a sense an average firm."[18] It had no special advantages or disadvantages and hence its costs of production reflected the average costs of the various firms within the industry. His analysis was based on a distinction between three time periods. First, in the "market period," supply was fixed and prices depended entirely on the strength of demand in relation to the fixed supply. Second, in the "short period," capital (or the productive capacity of a factory or other productive facilities) was fixed, but supply could be increased or decreased by altering the number

of workers laboring in conjunction with these productive facilities. Third, in the "long period," supply could be altered by increasing or decreasing both labor and capital. Any amount of productive facilities could be constructed in this period.

The market period was relatively unimportant in Marshall's analysis, so we will confine our discussion to the behavior of the firm in the short and long periods. Throughout the discussion, it was assumed that the firm had so many competitors that it could not itself directly influence either the prices that it had to pay for inputs or the price that it received for its output.

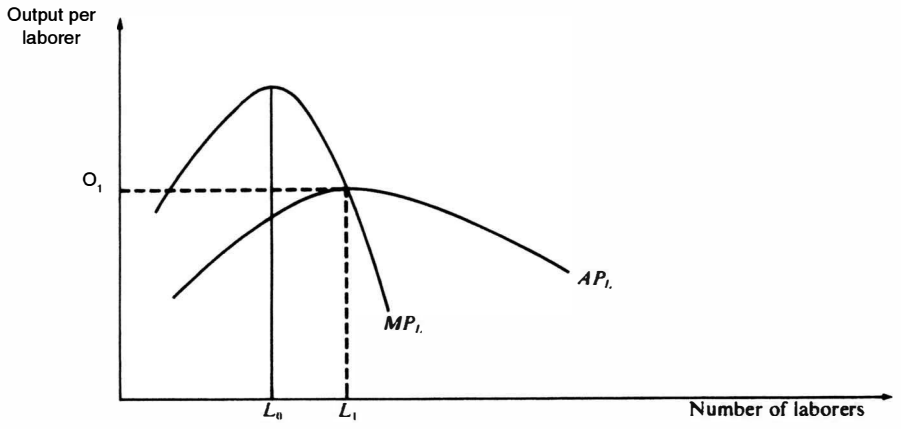
The Firm's Production and Cost Curves in the Short Period

In the short period, the size of the firm's productive facilities was fixed. It could therefore expand or contract output only by using a larger or smaller number of workers. But the factory (we will assume that the firm is a manufacturing firm) was constructed on the basis of a given technology of production. Therefore, average output per worker would be maximized when the firm hired the number of workers for which the factory was designed. The firm could alter the proportions of labor to capital (or the number of workers employed in the factory), but this alteration would affect the average productivity per worker. This meant that although labor and capital were substitutable, they were not *perfectly* substitutable. Therefore, as the capitalist hired more laborers, he or she would reach the point at which the factory was designed to function at optimum efficiency (that is, the point at which output per worker was maximized). Beyond that point, output per worker would decline.

Figure 11.1 illustrates the so-called law of variable proportions, or the law of diminishing returns to a variable factor of production.[19] The *marginal product of labor* (MP_L) line shows the increment to total output that results from hiring the last laborer at various levels of employment of labor within the factory. The *average product of labor* (AP_L) line shows the average output per worker at various levels of employment. If the factory starts from no production at all, the marginal product of labor will increase until the firm employs the number of laborers depicted by L_0 in the graph. Beyond this point the marginal product of labor declines as more labor is employed. As long as the MP_L *is* above the AP_L, the average productivity per laborer is increasing. When the factory hires the number of laborers depicted by L_1 on the graph, the average product of labor is maximized (at O_1 on the graph). If it hires more laborers than this, then the MP_L will be below the AP_L line, and the average product per laborer begins to decline.

When the firm computes the costs per unit of output at various levels of output, its cost curves appear to be nearly an inverse reflection of the product curves illustrated in Figure 11.1. As long as the average productivity per laborer is increasing, the amount of labor embodied in a unit of output is decreasing. Therefore, in Figure 11.1 the firm's *average variable cost* (or average labor

Figure 11.1 **Law of Variable Proportions**



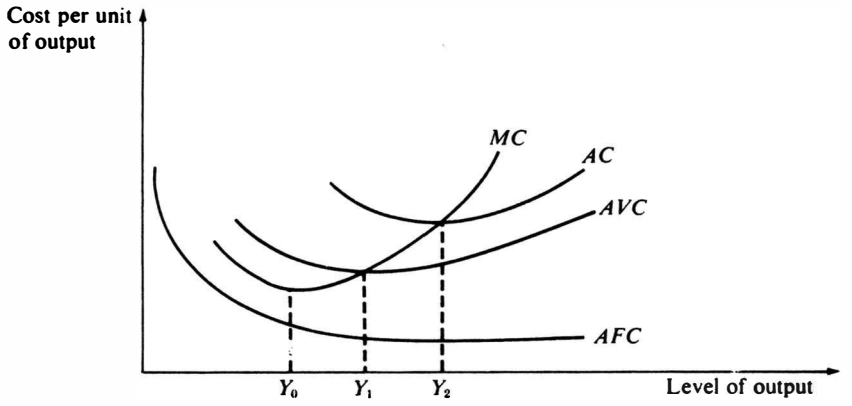
cost, insofar as we have assumed that labor is the only variable factor in the short period) per unit of output decreases continuously until it has hired L_1 number of laborers. As it hires laborers beyond that point, its average variable cost increases (reflecting the decrease in the average productivity of labor). Because the firm's capital is assumed to be fixed in the short period, its capital costs are fixed. As the firm hires more laborers and expands output, the *average fixed cost* per unit of output will decline continuously because the same level of total fixed costs is spread among ever larger quantities of output. The firm's *average cost is* simply the total of its average variable costs and average fixed costs.

The firm's cost curves are illustrated in Figure 11.2. Output level Y_1 corresponds to output per worker O_1 in Figure 11.1 (Y_1 is O_1 times the number of workers L_1 in Figure 11.1). At Y_1 average variable costs are minimized. At Y_2 average costs are minimized. Y_2 always involves a somewhat larger output than Y_1 because after average variable costs are minimized, there is a small range of output within which the declining average fixed costs more than offset the rising average variable costs. At outputs beyond Y_2, the rising average variable costs more than offset the declining average fixed costs, and, therefore, average costs rise continuously as output increases beyond Y_2. Marginal cost decreases continuously to output level Y_0, which corresponds to the output produced by L_0 number of laborers in Figure 11.1; it increases continuously beyond that point. The marginal cost curve intersects both the average cost and the average variable cost curves at their minimum point.

Equilibrium in the Short Period

The firm maximizes its profits by producing the level of output at which the price (as determined in the industrywide market) equals its rising marginal costs. If it produces at lower levels, then an increase in production will increase profits be-

Figure 11.2 **The Firm's Cost Curves**



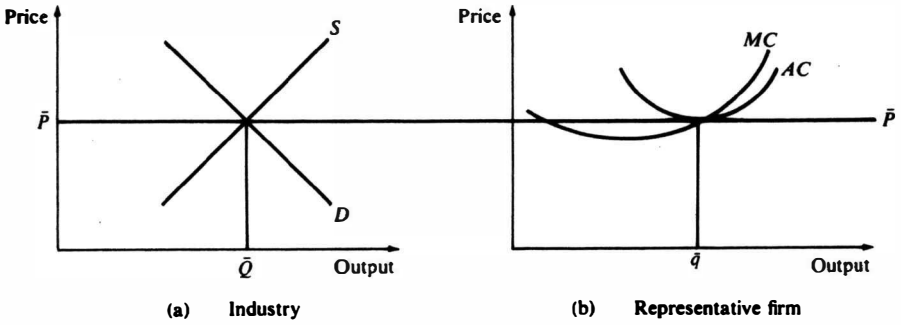
cause the price that it receives for the additional output will exceed the marginal cost of the additional output. If it produces beyond that level, its rising marginal cost of the additional output will exceed the price that it receives for the additional output. Therefore, the marginal cost curve shows the amounts a profit-maximizing competitive firm will supply at various prices and is the firm's *supply curve.*

Figure 11.3 illustrates the supply curve for an industry (which is the summation of the marginal cost curves of every firm in the industry) and the demand curve for the industry (which is determined by consumers' utility schedules). Part (a) of the figure shows the determination of the industry price by the intersection of the supply and demand curves. Part (b) shows the short-period equilibrium for the representative firm. The output for the industry is, of course, many times greater than the output for the representative firm. This is indicated in Figure 11.3 by using a capital Q for the industry's output and a small q for the firm's output, signifying the outputs are measured in different units (e.g., Q may be tons and q may be pounds of a product).

For the representative firm, the industry price equals its average cost. This does not mean that the representative firm earns no profit at equilibrium. Because neoclassical economists insist that a normal, or average, rate of profit on capital is a necessary part of the firm's capital costs, then such a profit would be included in average costs (as a part of its average fixed costs component). Any firm in this industry having higher costs than the representative firm will receive lower-than-normal profits. Any firm having lower costs than the representative firm will receive higher-than-normal profits, or excess profits.

Marshall believed that these latter firms have lower costs because they enjoy some natural advantage that other firms cannot acquire. He therefore called such excess profits "quasi-rents," and argued that they are similar to the "Ricardian rents" received by the owners of superior grades of land. In the long-period equilibrium, such quasi-rents will be incorporated into the firm's cost curves as ordinary rent costs, and every firm will ultimately receive only the normal rate of profit.

Figure 11.3 **Short-Period Equilibrium for an Industry and Its Representative Firm**



The Long Period and the Problem of Competition

In Marshall's long period, all productive factors could be varied, all costs were variable costs, and all quasi-rents disappeared. A long-period equilibrium, similar to the short-period equilibrium for a representative firm as illustrated in Figure 11.3, obtained for all competitive firms in all industries. The principal difference between the equilibrium pictured in Figure 11.3 and the long-period equilibrium was that in the latter, the size of the factory or general productive capacity of the firm had been adjusted to reflect an optimum of productive efficiency.

Marshall's equilibrium solution showed what proponents of the "invisible hand" had argued since Smith: competition not only equalized the rates of profit for all firms, it also minimized the costs of production (i.e., it maximized productive efficiency) and resulted in the consumer being able to buy all commodities at the lowest possible price—a price that just covered the socially necessary costs of production and yielded no surplus for any class to expropriate from any other class. Marshall did posit what he called both a "consumer's surplus" and a "producer's surplus." These concepts had nothing in common with the notion of an economic surplus found in the writings of Smith, Ricardo, and Marx, however. They were merely restatements of a theme common to all utilitarian economic theories. Specifically, they showed the benefits—in terms of utility—that all exchangers received over and above the utility that they would have had if they had been unable to exchange. Like his theory of the equilibrium of the firm, Marshall's surplus theory merely showed the "universal advantages" and social harmony he believed to be inherent in the process of competitive exchange.

The question with which Marshall had to deal in his long-period analysis was whether proportionate increases in all productive factors resulted in proportionate, less than proportionate, or more than proportionate increases in the firm's output. The commonsensical conclusion (as well as the conclusion of many economic theorists from earliest times to the present) is that

proportional increases in all inputs generally increase output proportionately (a situation defined as *constant returns to scale,* where average costs remain constant as the scale of productive operations expands). If such were the case, there would be no economic rationale (in terms of productive efficiency) for any particular sized firm in Marshall's long-period equilibrium. The emphasis on competition in the writings of the utility theorists would seem to suggest that they might favor legislated limits on the size of firms in order to maintain competition—particularly in view of the economic concentration that was developing in the late nineteenth century.

Marshall, however, did not believe that constant returns to scale would be the general rule. Following Ricardo, he believed that where land or natural resources were important in a production process, there was a tendency toward decreasing returns to scale (or increasing average costs over the long period). But when capital and labor were the primary factors of production, Marshall believed that there was a tendency toward increasing returns to scale (or decreasing average costs over the long period). He concluded that "while the part which nature plays in production shows a tendency to diminishing return, the part which man plays shows a tendency to increasing return."[20]

The increasing efficiency of greater size came from two sources. The first Marshall called "internal economies." Internal economies of scale were the result of improved organization of the firm: "An increase of labour and capital leads generally to improved organization, which increases the efficiency of the work of labour and capital."[21] The second source of increased efficiency Marshall called "external economies." External economies were due to benefits accruing to the firm (or the industry) from the output and pricing decisions of other firms (or industries). As examples of external economies of scale, Marshall discussed the benefits of industrial location and the benefits of closely interrelated secondary and tertiary industries.[22]

Marshall's notions of external economies (and the closely related notion of external diseconomies) were to become very important in neoclassical welfare economics, which we will discuss in Chapter 14. The importance of external economies in Marshall's analysis, however, was clearly stated in the following quotation:

> The general argument of the present book shows that an increase in the aggregate volume of production of anything will generally increase the size, and therefore the internal economies possessed by . . . [a] firm; that it will always increase the external economies to which the firm has access; and thus will enable it to manufacture at a less proportionate cost of labour and sacrifice than before.[23]

It would appear, then, that in the manufacturing sector of the economy, the greater efficiency of large-scale production would lead inevitably to monopolies and oligopolies. It therefore seems that Marshall's theory would have led him to one of three conclusions: First, he could have abandoned the general utilitarian argument (that the invisible hand of the competitive market harmo-

nized all interests and maximized the total social utility) and then defended capitalism on the basis of a new ideology stressing the social advantages of giant oligopolistic business corporations. Second, he could have argued that the social advantages of competition were more significant than the social advantages of the productive efficiency of large-scale production; thereby, he could have supported massive government intervention in the economy designed to break up large corporations and force them into perfectly competitive market structures where production inefficiencies were enforced by laws limiting the size of business firms. Third, he could have taken Marx's view that competition inevitably led to industrial concentration and that capitalist governments promoted rather than counteracted this tendency, and, therefore, advocated some form of socialism as the only possible means of taking advantage of the increased efficiency of large-scale production.

However, Marshall was unwilling to accept any one of these only possible practical conclusions. Like Mill, he wanted to retain the utilitarian ideology of the harmony of the invisible hand of the market. Also like Mill, he wanted to reject the notion (defended by Malthus and Senior, as well as most of the neoclassical economists within the more extreme laissez-faire utilitarian tradition) that the only hope for improving the plight of the poor was to promote the unfettered selfish quest for material gain by the rich. Not surprisingly, therefore, he proceeded in the same manner as Mill to incorporate into his intellectual system certain principles of moral and social philosophy that were quite incompatible with his utilitarianism.

Marshall's Ideological Defense of Capitalism

Marshall's ideological defense of capitalism was based on the incorporation of important elements of evolutionary social Darwinism into his theory. Unlike his American contemporary Thorstein Veblen, who was a thorough-going evolutionist, Marshall did not realize that the utilitarian social ethic was utterly incompatible with an evolutionary approach to economic theory.

Marshall set out "to consider the main bearings in economics of the law that the struggle for existence causes those organisms to multiply which are best fitted to derive benefit from the environment."[24] The principal conclusion he derived from his evolutionary approach was contained in the Latin phrase that appeared on the title page of the *Principles: "Natura non facit satum"* ("Nature contains no leaps"). All human progress, he argued, was very slow and proceeded only by minute *marginal* changes. Attempts to change society rapidly were doomed to fail and if undertaken would produce only misery. Social progress was generally a slow process of the hereditary improvement of a race or a nation. "This influence of heredity shows itself nowhere more markedly than in social organization. For that must necessarily be a slow growth, the product of many generations: it must be based on those customs and aptitudes of the great mass of people which are incapable of quick change."[25]

In this slow evolution of social institutions, a particular social structure might frequently appear to be exploitative on the surface. But the survival of that social structure for a long period of time was proof that, within its time and circumstances, its positive progressive features outweighed any of its defects:

> In early times . . . nearly all those nations which were leading the van of the world's progress were found to agree in having adopted a more or less strict system of caste: and this fact by itself proved that the distinction of castes was well suited to its environment, and that on the whole it strengthened the races or nations which adopted it. For since it was a controlling factor of life, the nations which adopted it could not have generally prevailed over others, if the influence exerted by it had not been in the main beneficial. Their pre-eminence proved not that it was free of defects, but that its excellences, relative to that particular stage of progress, outweighed its defects.[26]

The same argument, he believed, applied to modern capitalism. On the surface the capitalist system appeared as a "striking contrast" to the caste system. It also, however, offered "a no less striking resemblance to the system of caste."[27] While Marshall insisted that "the strength of . . . Marx's sympathies with suffering must always claim our respect,"[28] he nevertheless believed that these sympathies were misplaced: "The sacrifice of the individual to the exigencies of society as regards the production of material wealth seems in some respects to be a case of atavism, a reversion to conditions which prevailed in the far-away times of the rule of caste."[29] But in capitalism, as in the caste system, the benefits of the existing social structure far outweighed its defects.

Socialists had attacked the economic doctrines defending capitalism, but "the socialists did not study the doctrines which they attacked; and there was no difficulty in showing that they had not understood the nature and efficiency of the existing economic organization of society."[30] The socialists had not only wrongly perceived the economic doctrines that defended capitalism, they had wrongly perceived the motives of these economists:

> The fact is that nearly all the founders of modern economics were men of gentle and sympathetic temper, touched with the enthusiasm of humanity. . . . They were without exception devoted to the doctrine that the well-being of the whole people should be the ultimate goal of all private effort and all public policy. But they were strong in courage and caution; they appeared cold, because they would not assume the responsibility of advocating rapid advances on untried paths, for the safety of which the only guarantees offered were the confident hopes of men whose imaginations were eager, but not steadied by knowledge nor disciplined by hard thought. . . .
>
> It may be well therefore to note that the tendency of careful economic study is to base the rights of private property not on any abstract principle, but on the observation that in the past they have been inseparable from solid progress; and that therefore it is the part of responsible men to proceed cautiously and tentatively in abrogating or modifying even such rights as may seem to be inappropriate to the ideal conditions of social life.[31]

What was the most significant doctrine that the socialists had failed to understand and about which they should be "disciplined by hard thought?" It was none other than the invisible hand doctrine:

> If a man had a talent for managing business, he would be surely led to use that talent for the benefit of mankind: that meanwhile a like pursuit of their own interests would lead others to provide for his use such capital as he could turn to best account; and that his own interest would lead him so to arrange those in his employment that everyone should do the highest work of which he was capable, and no other; and that it would lead him to purchase and use all machinery and other aids to production, which could in his hands contribute more than the equivalent of their own cost towards supplying the wants of the world.
>
> This doctrine of natural organization contains more truth of the highest importance to humanity than almost any other which is equally likely to evade the comprehension of those who discuss grave social problems without adequate study: and it has a singular fascination for earnest and thoughtful minds.[32]

How then could "earnest and thoughtful minds" reconcile Marshall's belief in the greater efficiency of large-scale business firms with the perfect competition necessary for the working of the invisible hand? This was Marshall's most difficult problem, because he acknowledged that a capitalist owning an extremely large and growing firm "increases rapidly the advantages which he has over his competitors" and that this "process may go on . . . [until] he and one or two others like him would divide between them the whole of that branch of industry in which he is engaged."[33]

Once again Marshall was saved by his evolutionary theory of the life cycles of natural organisms. "Here," he insisted, "we may read a lesson from the young trees of the forest as they struggle upwards through the benumbing shade of their older rivals." Although these taller, better-established trees "have a better access to light and air than their rivals, they gradually lose vitality."[34]

As it was with trees, so it was with business firms:

> Nature still presses on the private business by limiting the length of the life of its original founders, and by limiting even more narrowly that part of their lives in which their faculties retain full vigour. And so, after a while, the guidance of the business falls into the hands of people with less energy and less creative genius if not with less active interest in its prosperity. . . . But it is likely to have lost so much of its elasticity and progressive force, that the advantages are no longer exclusively on its side in its competition with younger and smaller rivals.[35]

Thus, Marshall was able to salvage his faith in the permanence of perfect competition by believing that an industry was like a forest. Just as trees are constantly growing and dying, so with business; [a] firm's "decay in one direction is sure to be more than balanced by growth in another."[36]

The productive efficiency of giant firms also was not a force in preventing workers from becoming capitalists, in Marshall's opinion. In capitalism, he proclaimed, "the social relations of classes . . . are now perfectly variable and change their forms with the changing circumstances of the day."[37] In answer

to the socialist assertion that the separation of workers from the means of production resulted in the relative powerlessness of workers and the power of capitalists, he wrote:

> In speaking of the difficulty that a working man has in rising to a post in which he can turn his business ability to full account, the chief stress is commonly laid upon his want of capital: but this is not always his chief difficulty. . . . The real difficulty is to convince a sufficient number of those around them that they have the rare qualities [necessary to be a capitalist]. And the case is not very different when an individual endeavours to obtain from the ordinary sources the loan of the capital required to start him in business.[38]

The workers, if they had the appropriate moral virtues, could easily become capitalists. The existing English capitalists, or at least their forefathers, had achieved their position because they had "adopted a severe view of life; they took little delight in amusements that interrupted work, and they had a high standard as to those material comforts which could be obtained only by unremitting, hard work. They strove to produce things that had a solid and lasting utility."[39] Any worker with these character traits could still, he believed, become a capitalist.

But Marshall acknowledged that in its early phases capitalism was a harsh system for workers. In this period "free enterprise grew fast and fiercely, it was one-sided in its action and cruel to the poor."[40] Fortunately for the poor, Marshall reassured the reader, such cruelty was in the past. Capitalists, he believed, were becoming "chivalrous" and developing a benevolent concern for the poor.

> [Most social] evil may be lessened by a wider understanding of the social possibilities of economic chivalry. A devotion to public well-being on the part of the rich may do much, as enlightenment spreads, . . . in turning the resources of the rich to high account in the service of the poor, and may remove the worst evils of poverty from the land.[41]

Marshall cautioned "against the temptation to overstate the economic evils of our own age"[42] and against any impatience if economic chivalry took a long time to effect its social improvements. Social improvement ultimately depended on the elevation of human nature.

> But those elements of human nature which have developed during centuries of . . . sordid and gross pleasures, cannot be greatly changed in the course of a single generation.
>
> Now, as always, noble and eager schemers for the reorganization of society have painted beautiful pictures of life, as it might be under institutions which their imagination constructs easily. But it is an irresponsible imagination, in that it proceeds on the suppressed assumption that human nature will, under new institutions, quickly undergo changes such as cannot reasonably be expected in the course of a century, even under favourable conditions. If human nature could be thus ideally transformed, economic chivalry would dominate life even under the existing institutions of private

> property. And private property, the necessity for which doubtless reaches no deeper than the qualities of human nature, would become harmless at the same time that it became unnecessary.[43]

When Marshall spoke of some pleasures being "sordid and gross," he had of course completely abandoned the utilitarian premises underlying the invisible hand argument—an argument that he had asserted to be a "truth of the highest importance to humanity." In our previous discussions of the ideas of William Thompson and John Stuart Mill, we showed that social philosophies that discriminate among pleasures, calling some benevolent and lofty and others sordid and gross, contradict the very intellectual foundations of utilitarianism. We need not repeat that discussion here. Suffice it to say that this is another example of the general principle that a thinker's contradictions frequently furnish a revealing clue to his or her class orientation.

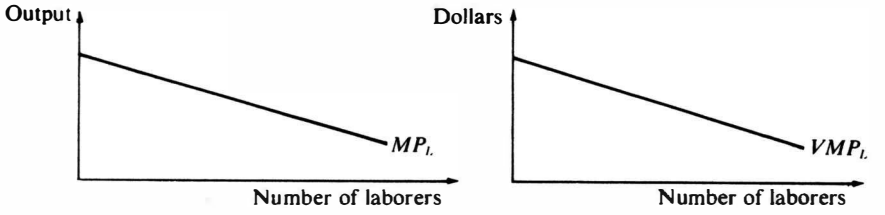
Clark and the Marginal Productivity Theory of Distribution

Earlier in this chapter, we made two important observations about neoclassical marginalism: first, when the principle of continuous marginal substitutability among the productive factors used in the production process is posited, the neoclassical theory of the firm becomes perfectly symmetrical analytically to the theory of household utility maximization. Second, given this factor substitutability, the theory of the firm's profit maximization can be viewed either from the standpoint of the revenue and expense per unit of output produced and sold or from the standpoint of the expenses of and the revenue resulting from the purchase and use of productive inputs. The latter standpoint represents the foundation of the neoclassical theory of income distribution.

While Marshall developed the neoclassical theory of the firm from the perspective of output, his development of distribution theory was inferior to that of John Bates Clark (1847–1938). This was because when Marshall analyzed the pricing of productive inputs, he assumed fixed technical coefficients of production and did not investigate the effects of marginal changes in the proportions in which the inputs were combined.[44] When he discussed the principle of substitution of productive factors, he analyzed only the substitution of different techniques of production, rather than investigating the effects of varying the proportion of the inputs used in a given production technique.[45]

For Clark, however, the principle of substitution of labor and capital was clearly developed. He argued that "in manufacturing and in transporting, too, the working force may often be varied perceptibly, with no change in the amount or in the character of the capital goods that are used in connection with it."[46] By holding the amount of capital constant and varying the quantity of labor, one could derive a schedule of the marginal product of labor for different levels of employment. From this schedule a graph showing the marginal product of labor, similar to Figure 11.1, could be constructed. If we assume

Figure 11.4 **The Firm's Marginal Product and Value of Marginal Product Curves**



(a) **Marginal product of labor** (b) **Value of marginal product**

that the marginal product of labor working with some fixed amount of capital declines continuously as more labor is added, then we get an MP_L curve as illustrated in part (a) of Figure 11.4.

In order to maximize his profits, the capitalist must know the *value of the marginal product of labor* and the price of labor. The value of the marginal product of labor in a competitive industry is merely the money value of the marginal product, or the marginal product times the selling price per unit. Thus, the VMP_L curve in part (b) of the figure is merely the MP_L curve from part (a) multiplied by the price of the output. The VMP_L tells the capitalist how much will be added to his revenues by hiring one more laborer at various levels of employment.

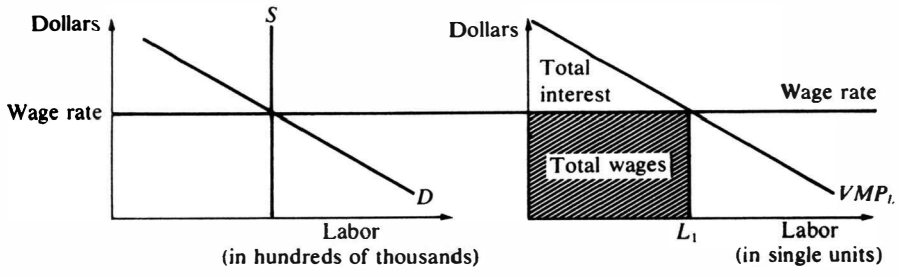
For a firm in a competitive industry, the price of labor is determined in the aggregate market for labor and that firm has no perceptible effect on either the price of labor or the price of the output that it sells. Figure 11.5 shows the labor market in part (a) and the level of employment of labor by a firm that will maximize the firm's profit, which occurs when the value of labor's marginal product is just equal to the wage rate as determined by the labor market. Therefore, at various wage rates, the VMP_L curve shows the corresponding amounts of labor that the firm will want to hire, that is, the VMP_L curve is the firm's *demand curve for labor.* Thus it follows that the aggregate demand for labor, shown in part (a) of Figure 11.5, is the summation of the VMP_L curves for all firms. The vertical supply curve for labor indicates that the supply of labor is, at any given time, fixed by the size of the labor force.

In Figure 11.5, the area of the shaded rectangle represents the total wages paid by the firm (the wage rate times the amount of labor hired, L_1) and the area of the triangle represents total interest (which is the residual of the value of the total product after wages are paid). The important point in this graph is that each firm hires labor until the value of labor's marginal product equals the wage. This is the necessary condition for the firm to maximize profits.

This was a very significant conclusion for Clark. The first page of the preface to *The Distribution of Wealth* began with this statement:

> It is the purpose of this work to show that the distribution of the income of society is controlled by a natural law, and that this law, if it worked without friction, would give to every agent of production the amount of wealth which that agent creates.

Figure 11.5 **Determination of the Wage Rate and the Firm's Level of Employment of Labor**



(a) Demand for and supply of labor in labor market

(b) Firm's profit-maximizing level of employment of labor, total wages paid, and interest received

> However wages may be adjusted by bargains freely made between individual men, the rates of pay that result from such transactions tend, it is here claimed, to equal that part of the product of industry which is traceable to the labor itself; and however interest may be adjusted by similarly free bargaining, it naturally tends to equal the fractional product that is separably traceable to capital. At the point in the economic system where titles to property originate—where labor and capital come into possession of the amounts that the state afterwards treats as their own—the social procedure is true to the principle on which the right of property rests. So far as it is not obstructed, it assigns to everyone what he has specifically created.[47]

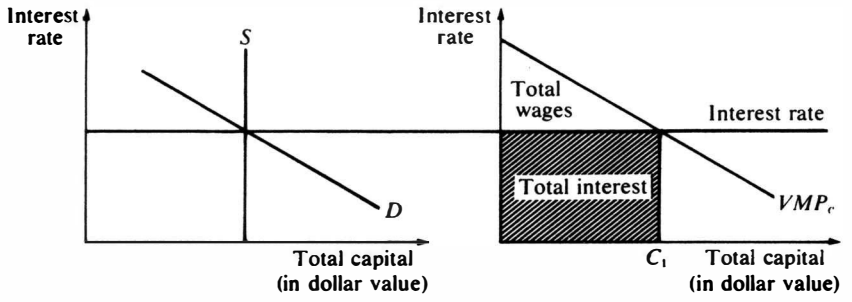
Thus, Clark had completed the task, originated by Say and Senior, of showing that the rewards to capitalists and workers were based on exactly the same principle. There was no surplus and there was no exploitation. Capitalists were rewarded by receiving what they created just as were laborers.

If in Figure 11.5 it appears that interest is a surplus, this is an illusion based on the fact that we held capital constant and varied labor to generate the MP_L curve. Because capital and labor are substitutable in neoclassical economic theory, we can just as easily hold labor constant and vary the amounts of capital used in production. Figure 11.6 is identical to Figure 11.5 except for the substitution of capital for labor as the variable productive factor.

It can be shown that in competitive equilibrium, if the total output involved in Figures 11.5 and 11.6 is the same, then the wage share and the capital share portrayed in both figures are the same. That is, the shaded rectangle representing total wages in Figure 11.5 represents exactly the same amount of wages as the triangle in Figure 11.6. The interest in both figures is also the same.

Each factor receives an income that is equal to the value of its marginal product. Although Clark merely assumed it, later economists rigorously proved that under the conditions of equilibrium in a perfectly competitive market, if each factor is paid the value of its marginal product, then these factor payments *exactly exhaust the value of the total output.* There is absolutely no possibility of exploitation. Each person receives the value of what his factor produces, and there is no surplus for anyone to expropriate.

Figure 11.6 **Determination of the Interest Rate and the Firm's Level of Employment of Capital**



(a) Demand for and supply of capital in the capital market

(b) Firm's profit-maximizing level of employment of capital, total interest received, and wages paid

Economics as Exchange and the Role of the Entrepreneur

In the exchange or utility perspective, profit disappears in equilibrium. Wages are the return to labor's contribution and interest is the return to capital's contribution. In the writings of the neoclassical economists, the distinction between classes—so important in the writings of Smith, Ricardo, Thompson, Hodgskin, Marx, and Mill—is completely eliminated. Most of the classical economists and Marx had defined profit as the residual surplus left after capitalists paid all the necessary costs of production. Neoclassical economists retained this definition of profit as a residual over and above costs. But in the neoclassical competitive equilibrium, *all income results from payments of the necessary costs of production. There is no residual; there is no surplus; and there is no profit.*

Neoclassical economics is the culmination of the tradition of Say, Senior, and Bastiat: *economics is exchange.* The entire economic process is seen in the following scenario: a population of egoistic, rational, utility-maximizing individuals, clustered in households, is originally endowed with inalienable private property rights to factors of production. To give up the use of these factors involves a sacrifice of utility. A strange minority of individuals has an assortment of talents and virtues qualifying them to be entrepreneurs. Entrepreneurs, as we will see, play the role of a fictional *deus ex machina* that performs a strange and contradictory—but absolutely necessary—function in the scheme.

Beginning with the initial endowments of productive factors, two sets of motives initiate a frenzy of rational, maximizing exchanges. First, households want to maximize utility. They do this by selling their factors and buying consumption goods to the point at which the ratio of marginal utility to price is the same for every factor that they sell and every commodity that they buy. Second, entrepreneurs want to maximize profits. They do this by buying factors

and selling consumption goods. They combine the factors, and, by virtue of a production function, these factors are transformed into consumption goods. The production function permits smooth, continuous substitution of the factors so that the marginal productivity of each factor is ascertainable.

The entrepreneur maximizes profits, in competitive equilibrium, when he or she pays each factor the value of its marginal product and sells each unit of the output at its cost. Because the factor payments exhaust the value of what is produced, the income accruing to the factor owners is exactly enough to purchase all of the consumption goods. And so the circular process goes on and on. Each factor owner sacrifices less total utility in selling his or her factors than he or she acquires through the purchase of consumption goods. Therefore, exchange benefits everyone, increases everyone's utility, and ensures a just and equitable harmony of interests. No one exploits anyone else. Each sells commodities and buys commodities at their equilibrium values. Each maximizes individual utility through the beneficence of the invisible hand of exchange. A beautiful symmetrical harmony, which Bastiat and Clark believed could only be explained by the divine benevolence of God, prevails everywhere.

But two very important *dei ex machina* are needed for the analysis (not to mention innumerable unrealistic assumptions about people, production functions, and economic institutions, which we will discuss in our chapter on neoclassical welfare economics). We saw both these *dei ex machina* in our discussion of Walras's ideas, and they recur in nearly all neoclassical analyses. The first is the entrepreneur whose ceaseless drive for maximum profits makes the system go, which results in the transformation of productive factors into consumption goods and thereby renders utility maximization possible.

If there ever was a modern-day counterpart to the myth of Sisyphus, it is the neoclassical myth of the entrepreneur. As we saw in Walras's theory, the entrepreneur is the organizer of production. As a classless entity, the entrepreneur may be a capitalist, a worker, or a landlord. If a capitalist, the entrepreneur receives only the interest on his or her capital that any other capitalist would receive. If a laborer or a landlord, the entrepreneur receives only the wage or rent that these same factors provide for a nonentrepreneur. There are, of course, real entrepreneurs in capitalism, but the entrepreneurs of neoclassical theory are purely mythical entities.

Such entrepreneurs are perpetually motivated by the quest for profit. But in the neoclassical vision of competitive equilibrium, there are no profits. So the neoclassical entrepreneur is perpetually scheming, worrying, buying, and selling in the quest for an illusory, chimerical will-o'-the-wisp. He or she never learns but continuously and assiduously pursues the Sisyphean task. In Clark's words:

> Normal prices are no-profit prices. They afford wages for all the labor that is involved in producing the goods, including the labor of superintending the mills, managing the finances, keeping the accounts, collecting the debts and doing all the work of directing the policy of the business. They afford, also, interest on all the capital that

> is used in the business, whether it is owned by the *entrepreneur* or borrowed from someone else. Beyond this there is no return, if prices stand exactly at their normal rate; and the reason for this is that *entrepreneurs* compete with each other in selling their goods, and so reduce prices to the no-net-profit level.[48]

The second *deus ex machina* is Walras's crier. Without this omniscient auctioneer, neoclassical economists have never been able to show how, in a general equilibrium setting, the free forces of supply and demand would ever automatically establish this set of equilibrium normal prices, in which each person gets just what his or her factors have created, and in which depression and involuntary unemployment are no longer matters of concern.

Thus, with initial endowments of private property secured as just and natural, equilibrium prices established by a cosmic crier, and mythical entrepreneurs constantly striving for nonexistent profits, the invisible hand of exchange maximizes every person's utility as they ubiquitously and universally engage in tens of thousands of marginal utility and price calculations and exchange their goods having less utility per dollar's value for goods having more utility per dollar's value. Everyone's interest is in harmony with every other person's interest, and each maximizes his or her own utility. Such is the way that capitalism functions according to the exchange or utility perspective. It is of no consequence in this theory that some exchangers ride to their plush thirtieth-floor offices in their chauffeur-driven Rolls-Royces, while others trudge, lunch pail in hand, to the bus stop, and still others trudge with no lunch pail to the unemployment line.

Clark's Defense of Private Property

The benevolence of the invisible hand ultimately rested on the belief that the existing laws of and distribution of private property were just and fair.[49] Clark believed that owning private property derived from instinctive behavior: "The land-owning instinct is the most effective motive in attracting laborers into the wealth-owning class."[50] The laws of private property reflected a kind of social ethos that embodied the people's moral feelings: "In free countries the people's sense of right expresses itself in laws; and in modern states it has actually pronounced in favor of the private ownership of land . . . and the Federal government, reflecting the will of the whole people, has confirmed it."[51]

Clark felt it to be particularly important to defend the private ownership of land, because he thought that the claims of socialists were most persuasive concerning ownership of land. Therefore, he believed that a defense of the private ownership of land would ultimately be a defense of all kinds of private ownership of productive factors. "Whatever logic there is in the case against landed property," he declared, "merges it in a case against all property, or in radical socialism."[52]

Unfortunately, in Clark's opinion, the majority of people were not aware of "the will of the whole people" or of the "people's sense of right." Because

of this ignorance, many people attacked the rights of private property: "It is to be expected that the assailants of private property should outnumber the defenders. This is usually true of an institution having great moral strength."[53] It was therefore of great importance that everyone understand the "natural law" that Clark had discovered: "The social procedure is true to the principle on which the right of property rests. . . . It assigns to everyone what he has specifically produced."[54]

Clark's Conception of Capital

The neoclassical marginal productivity theory of distribution explains the levels of wages and interest by positing a relationship between *physical quantities* of labor and capital used in production and corresponding *physical quantities* of marginal products attributable to these factors.

It is clear how one can aggregate physical quantities of labor. By defining one day's work by one laborer as a unit of labor, we can simply add the number of workdays to get the amount of labor expended in production. While there are differences among laborers, in most types of manufacturing these differences are quite unimportant:

> The men in an employer's services are thus interchangeable, it makes no difference to him which of them leaves his service. If the man who departs has been doing some kind of work that is quite necessary in conducting the business, the employer has only to put in his place the man who has been doing the work that is least needed.[55]

When entirely different skills and training are necessary, as we have seen in the chapters on Ricardo and Marx, the labor time spent in acquiring the necessary skills can be used to equate skilled labor to unskilled labor. Therefore, there are generally no insurmountable problems in summing physical quantities of labor.

It is an entirely different matter with capital. In summing the physical quantities of capital used in production, we must add together the most diverse set of objects. We must decide how, for example, to add screwdrivers to wheelbarrows to conveyor belts to blast furnaces to trucks and to innumerable other physical objects. The universal yardstick by which capitalists measure their capital is by its value. That is, each item is counted by adding its price to the prices of the other items constituting the stock of capital. While this method is perfectly appropriate for capitalists, who are only interested in making profit and not in providing an intellectual rationalization of profit making, it is absolutely inappropriate for the marginal productivity theory of distribution.

In neoclassical distribution theory, *the value of capital is determined by its productivity.* We cannot assume its value in order to ascertain its productivity. If we do this, then the productivity of capital will depend on its value. But in neoclassical theory, this line of causation must be reversed: the value of capital must depend on its productivity. Therefore, for the neoclassical marginal productivity theory of distribution to be consistent (i.e., for it to escape the

charge of intellectual circularity), there must be some method, completely independent of prices, to quantify the capital used in production.

Clark's treatment of this problem was most puzzling—and yet we will see that some neoclassical economists have continued to use Clark's formulation down to the present time. Clark insisted that his theory could not be understood unless one understood the distinction between capital and capital goods. "Capital consists of instruments of production," Clark wrote, "and these are always concrete and material."[56] When productive instruments were considered in their individual, material, concrete forms, they were called capital goods. Therefore, capital consisted of capital goods—and yet, at the same time, capital had qualities that were very different from the qualities of capital goods:

> The capital of the world is, as it were, one great tool in the hand of working humanity the armature with which humanity subdues and transforms the resisting elements of nature.
>
> The most distinctive single fact about what we have termed capital is the fact of permanence. It lasts; and it must last, if industry is to be successful. . . . Yet you must destroy *capital-goods* in order not to fail. . . .
>
> Capital-goods, then not only *may* go to destruction, but *must* be destroyed, if industry is to be successful; and they must do so, in order that capital may last. Seed-wheat must perish that wheat may abide.[57]

Another distinction was that "capital is perfectly mobile; but capital-goods are far from being so."[58] So capital was indestructible, perpetual, fluid, and mobile, while capital goods wore out and were concrete and immobile. Such a concept certainly is useful in the neoclassical theory of distribution but does not take us far in the quest to measure capital. Clark's example of wheat does not help. We can measure the quantity of wheat seed used up and compare it to our average inventories of wheat held over time. But wheat is wheat, and capital goods are incredibly diverse.

Clark answered that capital was an abstraction. It was "an abstract quantum of productive wealth, a permanent fund."[59] The word *quantum* certainly seems to name something quantifiable. But quantification is usually a concrete procedure directed at concrete things, not a quality of an abstraction. However, Clark never offered any definition of capital other than the following:

> A quantum of wealth, or a fund—if one of these be thought of apart from the concrete things that embody it, it is an abstraction; but if it be thought of as actually embodied in concrete things it is not an abstraction, but a material entity. . . .
>
> Guarding ourselves as carefully as we have done against the idea that capital ever lives in a disembodied state, we may safely use, for scientific purposes, the business man's formula. We think of capital as a sum of productive wealth, invested in material things which are perpetually shifting—which come and go continually—although the fund abides. Capital thus lives, as it were, by transmigration, taking itself out of one set of bodies and putting itself into another again and again.[60]

Clark appears to have been somewhat uncomfortable with his notion that abstract capital conceived of as "an abiding entity" could be measured.

"Productive power measured in units," he admitted, "is abstract."[61] However, he assured the reader that "such an abstract formula as this for describing a concrete thing is common in every sphere of thought."[62] If the reader had any final doubts, Clark offered one last profundity: "Life in itself is an abstraction."[63] He never offered the reader any method of quantitatively measuring either life itself as an abstraction or capital, which, as an abiding entity, is an abstraction.

But from Clark's time to the present, many neoclassical economists have continued to use the aggregate quantity of capital as a measurable physical quantity in their production (and later their growth) theories. They could not, with any theoretical consistency, use the value of capital as a means of measuring its quantity, because its quantity determined its marginal productivity, which, in turn, was supposed to determine its value. But insofar as "life in itself is an abstraction," they continued to construct theories as though they had found such a measure, when in fact they had not.

Böhm-Bawerk's Measure of Capital

Eugen von Böhm-Bawerk (1851–1914), who imagined that he had formulated the definitive critique of Marx, made an effort to formulate a measure of capital that would be independent of any prices. Böhm-Bawerk was aware of the inadequacies of most conservative economists' definition of capital, but he was most concerned to show the inadequacy of Marx's view of capital:

> Marx . . . would confine the conception of capital to those productive instruments which are to be found in the hands of persons other than the labourers themselves, and are used to exploit the labourers. With him, therefore, capital is the same thing as "means of exploitation." This distinction would be quite an important and suggestive one if the exploitation theory itself were correct. But since, as has been shown in my former work, it is not, the justification of the distinction based on that theory falls with it.[64]

Böhm-Bawerk was not content to criticize Marx's theory; he also developed a theory of capital and interest. He felt "obliged again to tread the heated path of controversy, in the hope that impartial and sober inquiry into the matter in dispute may succeed in ending it."[65] He was convinced he could show that interest was inevitable. "Those, then, who demand its abolition may . . . not, as the Socialists do at present, [base this demand] on the assertion that this kind of income is essentially unjustifiable."[66]

There were, Böhm-Bawerk asserted, only two "original" factors of production—land and labor. Capital came into existence as soon as it was realized that production takes *time*. Labor cannot be applied instantaneously but must be spread through time. Furthermore, people soon discovered that there are many alternative methods of producing a good. Some are short and direct and require relatively little time. Others are more "roundabout," more

indirect, and require more time. He believed that the lesson to be learned from this was obvious:

> It is—that a greater result is obtained by producing goods in roundabout ways than by producing them directly. Where a good can be produced in either way, we have the fact that, by the indirect way, a greater product can be got with equal labour, or the same product with less labour. . . .
>
> That roundabout methods lead to greater results than direct methods is one of the most important and fundamental propositions in the whole theory of production.[67]

With a given quantity of labor, Böhm-Bawerk believed that the longer the period of production, the greater would be the output of that labor, although diminishing returns would eventually develop. That is, if we continually increased the length of the production period by equal increments of time, production would continually increase but by smaller amounts with each successive increment.

It would seem, therefore, that if the only objective in producing were to maximize the amount produced, then people would always choose the longest possible production period. This was not so, Böhm-Bawerk argued, because just as people derived differing utilities from various goods in the present, so they derived differing utilities from having goods in the present and having goods in the future. That is, *people had time preferences.* Moreover, nearly all people preferred goods in the present to goods in the future. Böhm-Bawerk felt that there were several reasons for this preference. First, there were "cases of immediate distress and necessity,"[68] in which the present seemed more important than the future. Second, people had a tendency to "underestimate future pleasures" because of "the incompleteness of imaginations," or "a defect in the will," or "the shortness and uncertainty of life."[69]

It followed that the time pattern of production affected people's utilities in two separate and opposite ways. First, more goods would yield more utility, and the longer the period of production the more goods would ultimately be available for consumption. Second, the sooner people got their goods the more utility they would expect, and a longer period of production meant a longer postponement of consumption.

Now if we began with a very short production period with low labor productivity, increasing the time period would probably be beneficial (that is, increase utility) because the increased utility of more goods would outweigh the disutility of waiting slightly longer for the goods. But as we continued adding equal time increments to the production process, two things would happen. First, the marginal increment to output would continuously decline. Second, the disutility of a marginal increment of waiting time would increase. It is obvious that at some point we would have extended the production period to the length at which the utility of the marginal increment in output would just equal the disutility of the marginal increment in waiting time. At that point, society's utility is maximized over time.

In this conception, the quantity of capital is an index of three separate

aspects of the production process: first, the amount of the original factors of production used; second, the length of the production period; and third, the temporal pattern of the usage of the original factors of production throughout the period of production. To lengthen the production process is to increase the amount of capital. In equilibrium, each person's time preference is given by the ratio MU_t / MU_{t+1} (where the subscripts t and $t + 1$ refer to two successive time periods). This ratio is also equal to the marginal productivity of capital (or the ratio of the value of inputs in time period $t + 1$ over their value in time period t). Both these ratios are equal to (and, in fact, they are the determinants of the equilibrium value of) one plus the interest rate.

Thus, in Böhm-Bawerk's analysis, the definition of the nature of capital was completely subsumed in the process of utility maximization simply by introducing time into the analysis. Böhm-Bawerk appeared to have provided a solution to Clark's problem of measuring capital, because the quantities of the original factors used in production, the length of the production period, and the time pattern of input usage could be ascertained independently of any prices. We will see in Chapter 16, however, that this measure of capital is totally unsatisfactory because it is an index of several numbers, and, under certain conditions, the numbers yield indexes that contradict the premises of the marginal productivity theory of distribution. We will see that, although neoclassical economics has yet to find a consistent method of measuring capital that is independent of prices, the ideological value of the marginal productivity distribution theory is so great that the majority of neoclassical economists continue to defend the theory.

Capitalist Class Relations in Neoclassical Distribution Theory

Despite Böhm-Bawerk's insistence that his theory was completely different from that of Senior, it appears to be merely an extension and elaboration of the latter's theory. Böhm-Bawerk identified capitalist production with roundabout production. He thereby assumed that any production process that uses tools and intermediate products was a capitalist production process. The following quotation summarizes his view of the essential difference between capitalists and laborers:

> The adoption of capitalist methods of production is followed by two consequences, equally characteristic and significant. One is an advantage, the other a disadvantage. The advantage . . . consists in the greater technical productiveness of those methods. . . .
>
> The disadvantage connected with the capitalist method of production is its sacrifice of time. The roundabout ways of capital are fruitful but long. . . . In the overwhelming majority of cases we must tread the roundabout ways of capitalist production under technical conditions of such a nature that we have to wait, and often for a very long time, before we get the ripe final product. . . . In the loss of time which is . . . bound up with the capitalist process lies the sole ground of that much-talked-of and much-deplored dependence of labourer on capitalist. . . . It is only because the

> labourers cannot wait till the roundaboutness . . . delivers up its products ready for consumption, that they become economically dependent on the capitalists.[70]

In other words, only a difference in mental and emotional attitudes differentiated laborers from capitalists. They had different time preferences. Laborers wanted their reward now, while capitalists had the moral character that permitted them to wait. But Böhm-Bawerk's notion of waiting, as well as Marshall's notion of waiting, was essentially identical to Senior's notion of abstaining. Like Marshall, Böhm-Bawerk was somewhat embarrassed by the use of the word *abstinence,* because he was uncomfortable with Marx's critique of Senior's claim that abstinence was a unique social cost of production. Böhm-Bawerk quoted the following passage from Marx:

> It has never occurred to the vulgar economist to make the simple reflection, that every human action may be viewed as "abstinence" from its opposite. Eating is abstinence from fasting, walking abstinence from standing still, working abstinence from idling, idling abstinence from working, etc. These gentlemen would do well to ponder, once in a while, over Spinoza's *Determinatio est Negatio.*[71]

Böhm-Bawerk simply dismissed Marx's point by stating that "to my mind there is more dialectic than truth in this argument."[72]

We will conclude this chapter by repeating our earlier assertion. Neoclassical economics sees the economic process as simply a series of exchanges in which equivalent is exchanged for equivalent. Everyone benefits because, while each person gets exactly the value equivalent of what he or she gives up, each person derives more utility from what he or she gets than from what he or she gives up. The market therefore harmonizes everyone's interest and maximizes everyone's utility. There are no classes or class antagonisms in this theory. Some exchangers have a higher moral character than others (i.e., some practice abstinence and some do not), but each person receives, through exchange, the value equivalent of what his or her factors create. The marginal productivity theory of distribution is an integral part of the general neoclassical theory of exchange.

In discussing his distribution theory, Marshall expressed a conclusion that was shared by all three of the theorists discussed in this chapter (as well as most of the subsequent neoclassical theorists). After discussing his theory of input pricing (or income distribution), he concluded: "This statement is in close harmony with such common sayings of every-day life, as that 'everything tends to find its own level,' that 'most men earn just about what they are worth.'"[73]

While Marshall's statement summarized the moral conclusion inherent in the marginal productivity theory of distribution, the following quotation from Clark's *The Distribution of Wealth* most clearly illustrates the central concern that has motivated neoclassical developers of the marginal productivity theory of distribution from his own time to the present:

> The welfare of the laboring classes depends on whether they get much or little; but their attitude toward the other classes—and, therefore, the stability of the social state—depends chiefly on the question, whether the amount that they get, be it large or small, is what they produce. If they create a small amount of wealth and get the whole of it, they may not seek to revolutionize society; but if it were to appear that they produce an ample amount and get only a part of it, many of them would become revolutionists, and all would have the right to do so. The indictment that hangs over society is that of "exploiting labor." "Workmen," it is said, "are regularly robbed of what they produce. This is done within the forms of law, and by the natural working of competition." If this charge were proved, every right minded man should become a socialist; and his zeal in transforming the industrial system would then measure and express his sense of justice.[74]

Notes to Chapter 11

1. Alfred Marshall, *Principles of Economics,* 8th ed. (London: Macmillan, 1961), p. 79.
2. Ibid., p. 80.
3. Ibid., p. 83.
4. Ibid., pp. 86–97.
5. Ibid., pp. 103–9.
6. Ibid., pp. 109–14.
7. For example, M. Blaug, *Economic Theory in Retrospect,* rev. ed. (Homewood, IL: Irwin, 1968), pp. 327–430.
8. Marshall, *Principles,* pp. 120–43.
9. Ibid., p. 117.
10. Ibid.
11. Ibid., p. 193.
12. Ibid.
13. Ibid., p. 99.
14. Ibid., p. 100.
15. Ibid., pp. 336–37.
16. Ibid., p. 335.
17. Ibid., p. 337.
18. Ibid., p. 265.
19. The remainder of this section and some of the sections describing Clark's distribution theory and Böhm-Bawerk's capital theory are written in the present tense because these analyses have become standard in contemporary elementary economics textbooks.
20. Marshall, *Principles,* p. 265.
21. Ibid.
22. Ibid., pp. 222–31.
23. Ibid., p. 265.
24. Ibid., p. 201.
25. Ibid., p. 203.
26. Ibid., pp. 203–4.
27. Ibid., p. 204.
28. Ibid., p. 488.
29. Ibid., p. 204.
30. Ibid., p. 631.
31. Ibid., pp. 39–40.
32. Ibid.
33. Ibid., p. 263.
34. Ibid.
35. Ibid., pp. 263–64.
36. Ibid., p. 264.
37. Ibid., p. 204.

38. Ibid., p. 257.
39. Ibid.
40. Ibid., p. 615.
41. Ibid., p. 599.
42. Ibid., p. 600.
43. Ibid.
44. Ibid., p. 318.
45. Ibid., pp. 335–37.
46. John Bates Clark, *The Distribution of Wealth* (New York: Augustus M. Kelley, 1965), p. 101.
47. Ibid., p. v.
48. Ibid., p. 111.
49. For an excellent discussion of this and many other aspects of Clark's writings, see John F. Henry, "John Bates Clark and the Origins of Neoclassical Economics," Ph.D. diss., McGill University, 1974.
50. John Bates Clark, "The Ethics of Land Tenure," *International Journal of Ethics* (October 1890): 69.
51. Ibid., p. 62.
52. Ibid., p. 77.
53. Ibid., p. 62.
54. Clark, *Distribution of Wealth,* p. v.
55. Ibid., p. 103.
56. Ibid., p. 116.
57. Ibid., p. 117.
58. Ibid., p. 118.
59. Ibid., p. 119.
60. Ibid., pp. 119–20.
61. Ibid., p. 121.
62. Ibid.
63. Ibid.
64. Eugen von Böhm-Bawerk, *The Positive Theory of Capital* (New York: Stechert, 1923), p. 57.
65. Ibid., p. 23.
66. Ibid., p. 364.
67. Ibid., pp. 19–20.
68. Ibid., p. 249.
69. Ibid., pp. 254–55.
70. Ibid., pp. 82–83.
71. Ibid., p. 123.
72. Ibid.
73. Marshall, *Principles,* p. 335.
74. Clark, *Distribution of Wealth,* p. 4.